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Dr.Enas El-Sharawy

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This Software Design Specification was prepared and provided as a deliverable for [Course Name, number, term], and it will be used by [name of end user].

This document is based in part on the IEEE Recommended Practice for Software Design Descriptions.

#	Name	ID	Role
1	Khawla Hani Almoalem	2230003157	Leader
2	Fatimah Moayed Alawa	2230005350	Member
3	Zainab Esaeed	2230000664	Member
4	Zahra Aiman Alsadah	2230004418	Member
5	Jood Mohammed Alzayer	2230002118	Member
6	Fatimah Majed Almur	2230004016	Member
7	Remas Nasser Alqahtani	2230002281	Member

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1. Introduction

This Software Design Specification (SDS) describes the overall architectural design, component design, and implementation design for the Sanad platform, a holistic digital accessibility system designed to allow an impressive level of independence, activity, and self-determination for individuals with disabilities. This document will be used as a guide by developers to assist in the development of a system that meets both functional and non-functional requirements from the Software Requirements Specification (SRS). The document describes the overall design decisions, the architecture of the system, data models, user interfaces, integration points, and issues pertaining to accessibility, security, and scalability. The document has an overview section discussing the purpose, scope, intended audience, documentation references, key definitions, and a summary of contents, followed by the design specification sections that detail the design.

1.1 Purpose

- The main purpose of this Software Design Specification (SDS) is to provide a comprehensive design specification of the Sanad platform, transitioning the ability and features from the SRS to an implementable architecture and technology.
- This document seeks to establish a framework that will guide the developer(s) on building a secure, accessible, and scalable platform for those with disabilities using interactive maps, community forums, booking of services, and libraries of assistive technology.
- This document also provides continuity between design and requirements, assists code implementation, testing, and maintenance and serves as a framework towards the accomplishment of standards, such as WCAG 2.1 (Web Content Accessibility Guidelines), and the Saudi Regulations of the Protection of Personal Data protection legislation.

1.2 Scope

1. The scope of this Software Design Specification includes the design of the web and mobile applications of the Sanad platform, along with the backend services, databases, APIs, and applicable external systems (e.g., Google Maps, payment platforms).
2. It includes the architecture, component description, data model, user interface design, and security protocols, all while focusing on the core functionality of user management, maps, community forums, service bookings, learning and employment resources, and assistive technologies.
3. - In-scope: Web-based application of users with or without barriers, mobile application for users with barriers, connections to backend services, various service locations integrated with Google Maps and other needed systems, and appropriate payment services.
4. Out-of-scope: Any designs related to hardware ecosystems (e.g., details on user hardware design), implementation on third-party vendor's specific systems (e.g., user interaction with workable Google Maps code), or post-launch operations (e.g., user training, advertisements).

5. The scope applies specifically to Version 1.0 of the Sanad platform but can expand in subsequent versions based on the organization's change management processes.

1.3 Definitions, Acronyms, and Abbreviations

Terminologies are listed in the table below:

Terminology	Definitions
Sanad	Sanad Platform is a unified digital system that provides accessible digital services to empower people with impairments.
Accessibility	The process of designing products, devices, services, or environments that can be accessed and used by people with disabilities.
Architectural Design	Architectural design generally refers to the high-level design of a software system along with how the various components of the system interact with each other.
Assistive Technology	Assistive technology (AT) may refer to some tools or devices that support people with disabilities in performing tasks.
Backend services	Backend services describe the server-side components of a software system.
Business logic	Business logic includes the rules and processes that determine how data is processed or how the software system behaves.
The client-server model	The client-server model, in general, describes a model of a distributed application where the client (web/mobile app) requests services from the server (backend).
Component design	Component design generally refers to the detailed design (i.e., the structure, state, behavior, etc.) of each of the components of a software program.
Data Flow	The route data follows across a system, from being input, processed, and output.
Data Model	The representation of the type data structures and how they relate to each other in a system.
Digital Accessibility	Providing access to digital tools and platforms for people with disabilities.
Frontend	The end user or client facing portion of a software application or engagement with it, including the user interface and user interactions.
Functional Requirements	Details on technical specifications of what the system has to do for functionality - i.e. user registration, map engagements, etc. Derived from the SRS, and answered in the SDS.
Implementation Design	The detailed implementation plan, including choice of technology stack hired and use of other services and systems.
Integration Points	The locations where the system connects to, or is related to, an external service or system.
Layered Architecture	Layers were through and functional application of design principles and patterns creating the software architecture for layering.
Microservices	A type of software architecture in which applications are built as a set of small, independent services.
Modular System Architecture	A design paradigm in which the system is partitioned into independent modules, to facilitate activity and maintenance.
Non-Functional Requirements	The nonfunctional specifications of how the system performs (e.g., security, scalability, accessibility).

RESTful APIs	APIs that adhere to principles of REST for communication between layers of the system.
Scalability	The ability of a system to accommodate the growth of users or data.
Security Protocols	An actions taken by a system that delivers protection against a threat.
Self-Determination	The ability of individuals to choose, make decisions, and have control over their lives.
Social Inclusion	The process of improving the conditions under which individuals can participate in society, derived form the WHO
Technology Stack	The combination of technologies utilized to aid in the construction of a system for consumers.
User Interface (UI) Design	Within computing, UI design pertains to the real and/or visual and interactive design elements that consumers engage with.
WebSocket	A network communication protocol for real-time bidirectional data exchange.
Fields	An attribute or column in a table that stores some specific data about that row with an associated type and rules.
Constraints	A rule that defines which values can be used in a field, limiting the values to ensure validity and referential integrity.
Entity	An attribute or column in a table that stores some specific data about that row with an associated type and rules.
Inline Validation Message	A message displayed next to a form filed immediately after invalid input to guide the user.
Opportunity Card	A card that shows a specific opportunity, such as a new job or service.
Modality	The method of delivering a service, either remotely or on site.

Table 1- Definition

Acronyms	Meanings
SDS	Software Design Specification
SRS	Software Requirements Specification a document outlining the software's objectives, features, and limitations.
SQL	Structured Query Language
2FA	Two-Factor Authentication (a security process that enables users to provide two separate forms of verification to validate their identity).
API	Application Programming Interface (a collection of rules and protocols to access a software application or web-based tool). AWS: Amazon Web Services (a cloud computing platform offered by Amazon).
CRPD	Convention on the Rights of Persons with Disabilities (a United Nations treaty to protect the rights of persons with disabilities).
DGA	Digital Government Authority (a Saudi Arabian government agency focused on digital transformation and accessibility).
Flutter	Flutter is a user interface toolkit for building applications which are compiled natively (a mobile app development open-source framework by Google).

HTML5	HTML5 or Hypertext Markup Language 5 is the latest version of the markup language for web pages.
HTTPS	HTTPS, or Hypertext Transfer Protocol Secure is a secure version of HTTP designed for secure communication over computer networks.
IEC	IEC, or International Electrotechnical Commission, is a standardization organization in the field of electrotechnology. IEEE: IEEE, Institute of Electrical and Electronics Engineers, is an organization of professionals that advances technology for humanity in the field of electrical engineering and computing.
ISO	ISO stands for International Organization for Standardization and is the international standard-development organization.
JWT	JSON Web Token (a compact, URL-safe method for representing claims to be transferred between two parties).
MySQL	My Structured Query Language (an open-source relational database management system).
Node.js	A JavaScript runtime built on Chrome's V8 JavaScript engine (a server-side development platform).
React	A JavaScript library for building user interfaces (a library for creating interactive web user interfaces).
Redis	Remote Dictionary Server (an in-memory data structure store used as a database, cache, and message broker).
REST	Representational State Transfer (an architectural style for designing networked applications, typically used for APIs).
UI	User Interface (the space where human and machine interactions occur).
UN	United Nations (an international organization that was formed in 1945 to promote peace and cooperation).
W3C	World Wide Web Consortium (a worldwide community that focuses on building open standards for the Web)..
WCAG	Web Content Accessibility Guidelines (guidelines for improving the accessibility of web content for people with disabilities).
WHO	World Health Organization (a United Nations agency that specializes in international public health).
XSS	Cross-Site Scripting (a sort of vulnerability seen in web applications).
AT	Assistive technology may refer to some tools or devices that support people with disabilities in performing tasks
etc.	abbreviation for the Latin phrase "et cetera," which means "and other things" or "and the rest". It is used at the end of a list to indicate that there are more similar items that could be included but are not explicitly mentioned.
Pk	Primary key a unique value for every row of a table, preventing duplicate rows and serving as a reference.
Fk	Foreign key a field that references the primary key of another table, which allows us to establish relationships between sets of data.

Table 2-Acronyms

1.4 References

- [1] Software requirements specification (SRS) for the Sanad platform (version 1.0).
- [2] IEEE Std 830-1998, IEEE recommended practice for software requirements specifications, Institute of Electrical and Electronics Engineers, 1998.
- [3] ISO/IEC/IEEE 29148:2018, Systems and software engineering - life cycle processes - requirements engineering, International Organization for Standardization, 2018.
- [4] World Wide Web Consortium (W3C), Web content accessibility guidelines (WCAG) 2.1, W3C recommendation 2018.
- [5] Digital Government Authority (DGA), Digital accessibility guidelines, Riyadh: Digital Government Authority, 2022.
- [6] World Health Organization (WHO), World report on disability, Geneva: World Health Organization, 2011.
- [7] United Nations, Convention on the rights of persons with disabilities (CRPD), New York: United Nations, 2006.
- [8] Test plan for the Sanad platform.
- [9] Development plan for the Sanad platform.
- [10] User manual for the Sanad platform.

2. System overview

The Sanad platform is a mobile and web-based system that provides people with disabilities with active support in their daily living, social participation/engagement, education, work, and community services. It is developed as an independent application with external integrations and follows a modular system architecture for scalability and maintainability. The system is built on a client-server model, using a React-based web interface, a Flutter/React Native mobile app and a Node.js backend with a MySQL database. Additionally, accessibility (WCAG 2.1 compliance), security (HTTPS and encryption), and performance (e.g., caching, lazy loading) are key design principles. Background: The platform allows individuals with disabilities to leverage a digital tool designed to address contemporary gaps in digital accessibility, informed by research on assistive technologies and social inclusion, all included in the SRS.

2.1 System Features

- User Management: Manages registration, authentication (with 2FA), and user profile settings based on individual disability types.
- Interactive Map: Shows accessible locations and other filters, ratings, and user submissions, using the Google Maps API.
- Community Forum: Supports posts/replies and moderation with an area to share experiences and uplifting stories.
- Service Bookings: Allows the user to book an interpreter or volunteer for help, tracking in real-time with notifications.
- Learning and Employment: Searching for courses and jobs with external links and favorite saved.
- Assistive Technology Library: Compiles cataloged with reviews, check for compatibility for individual use and purchase links.
- Admin Dashboard: Monitors platform activity, moderation of content, and analytics.

2.2 Design Strategy and Structure

- Architecture: Our system utilizes a layered architecture consisting of presentation (UI), application (business logic), and data (database) layers. It utilizes RESTful APIs for communication between each layer and microservices to provide modularity to the system.
- Technology Stack: The technology stack includes a frontend using (React/HTML5), a backend with (Node.js/Express), a single database using (MySQL), mobile applications using (Flutter), and system integrations where needed (Google Maps or PayPal).
- Security Architecture: Security considerations include: JSON Web Token (JWT) for authentication, end-to-end encryption for video calling, and validated input to prevent XSS and other injections.
- Accessibility Design: Accessibility design practices ensure high-contrast modes, screen reader compatibility, and keyboard navigation is available on all interfaces.
- Scalability: Infrastructure is created to scale with caching technologies (Redis), load balancing, and deployment to the cloud (AWS) for supporting up to 10,000 users concurrently.
- Data Flow: to support a dynamic user experience whenever a user makes an input this will trigger a REST API call to the backend. This is where business logic will attach web socket events while the backend retrieves data and records data to a database. Web socket publishes the data in real-time along with configuration settings.

3. Design Considerations

3.1 Assumptions and Dependencies

The design of the **Sanad System** is based on a set of assumptions and dependencies that are critical to the successful implementation and operation of the software. These assumptions guide development decisions and define the operational environment:

- **Reliable Internet Connectivity:**
Users are assumed to have continuous and stable internet access (4G/5G or Wi-Fi) to utilize the system's primary features, including the interactive map, forum, e-learning modules, and appointment scheduling.
- **Modern User Devices and Browsers:**
The system is intended to run on modern mobile devices and web browsers. Supported environments include iOS 12.4.1 or higher, Android Oreo 8.1 or higher, and browsers compatible with HTML5, CSS3, and ES6 standards.
- **Third-Party Services and APIs:**
The system depends on external APIs, such as Google Maps, cloud storage (AWS S3), push notifications (Firebase), and payment gateways. Stability, uptime, and service-level agreements of these services are critical to system functionality.
- **Regulatory Compliance:**
The software is developed under the assumption of compliance with the Saudi Personal Data Protection Law (PDPL) and internationally recognized privacy standards such as GDPR.
- **Accessibility Guidelines Compliance:**
All system interfaces and functionalities are assumed to follow **WCAG 2.1 AA** accessibility standards to ensure usability for people with varying disabilities.
- **Operational and Maintenance Resources:**
Continuous availability of development, IT support, and administrative personnel is assumed for system maintenance, updates, and troubleshooting.
- **Scalability Expectations:**
The system assumes moderate initial load, with potential growth to accommodate up to 10,000 concurrent users.
- **Authentication and Security Mechanisms:**
Authentication services rely on JWT-based tokens and optional OAuth2 providers (Google, Apple). The assumption is that these mechanisms are stable and secure.
- **Content and Data Availability:**
It is assumed that initial content, such as location data, service providers, and assistive technology libraries, will be provided by system administrators.
- **Assumption: Consistent Backend Infrastructure**

The system assumes that a stable backend with a stable architecture (such a layered architecture or MVC) is able to process requests deterministically to ensure reliable performance, maintainability and concern separation across system components.

- **Dependency: Database Availability**

The system depends on a MySQL / cloud-hosted database to remain continually available. Any downtime or dysfunction of the database will impact core functionalities, including authentication, map services, appointments, and user actions.

- **Dependency: Authentication Services**

The authentication mechanism will be dependent on reliable JWT-based token generation and validation. The optional OAuth2 identity providers (Google and Apple) will also need to work reliably to alternate login pathways.

- **Dependency: Push Notification Delivery Service**

The system is reliant on Firebase Cloud Messaging (FCM) to deliver real-time notifications to site users, including appointment updates, volunteer requests, or site messages. Any downtime or malfunction to the FCM will disrupt notification delivery in a timely matter.

3.2 General Constraints

The following global constraints were considered during the design of the Sanad System. These constraints influence architecture, functionality, and user experience:

- **Hardware and Software Environment:**

The system is optimized for modern mobile devices and browsers, with stable internet connections. Data storage and retrieval are performed through secure centralized databases to maintain consistency and accessibility.

- **Interface Requirements:**

The interface is designed for accessibility and ease of use. Input fields, error messages, pop-up notifications, standardized fonts, icons, and images are clearly defined to support navigation and comprehension by all users.

- **Data Repository and Distribution:**

User data, including personal information, appointments, and activity logs, are securely stored in a centralized database. Access is role-based, ensuring that users can only retrieve data according to their permissions.

- **Security and Protection:**

Sensitive information such as user IDs, passwords, and OTPs is encrypted in transit and at rest. Access to system functionalities is restricted based on user roles to protect confidentiality, integrity, and system reliability.

- **Performance Requirements:**

The system is designed to handle expected user load efficiently. Optimization includes database indexing, caching, content compression, load balancing, and offline support for intermittent network connectivity.

- **Network Communications:**

The system depends on reliable network communication for real-time synchronization, notifications, and API calls. Offline caching ensures continuity in case of temporary network issues.

- **Verification and Validation (Testing):**

Comprehensive testing is conducted according to industry best practices, including functional, integration, performance, security, and accessibility testing. Both automated and manual tests ensure compliance with requirements and regulatory standards.

- **Standards Compliance and Interoperability:**

The system adheres to **WCAG 2.1 AA** accessibility guidelines and supports integration with third-party APIs and cloud services to maintain consistent functionality across platforms.

- **Resource Constraints:**

Budget, timeline, and availability of domain specialists, including accessibility experts, influence the design. A phased deployment approach is implemented to prioritize essential functionalities while maintaining quality standards.

- **Architectural Limitations :**

It is required that the system consist of a multi layered architecture wherein the presentation layer, business logic layer and data access layer are separated. This will allow for scalability, maintainability and a clear separation of responsibilities in the system.

- **Technology stack constraints :** It has been determined that the design will be limited by the technology stack that was predetermined for the Sanad Platform. They include:

- Frontend: React for the web and Flutter for mobile applications.
- Backend: Node.js with Express.js.
- Database: MySQL.
- Cloud storage: AWS S3 for hosting media and files.

In order to accommodate these technologies, design constraints will affect patterns of integration, deployment environments and performance optimization strategies, and used together these constraints and assumptions provide: a unified framework for building a secure, reliable and accessible platform to support participation, independence and social inclusion for people with disabilities.

4. User Interface Design

This section provides an overview of the user interface of the Sanad mobile application from the user's perspective. It explains how a user moves between the main functions, executes core tasks, and receives feedback from the system.

4.1 Overview of User Interface

The Sanad mobile application uses a simple card-based layout with a persistent bottom navigation bar in its design to make it more accessible and easier to learn by users with different abilities. Once the user has been authenticated, the five main screens that users will interact with are accessible from:

- Home: application entry point. It contains a search bar, quick access controls (High Contrast, Text Size and Text-to-Speech), featured accessible places and recent forum posts.
- Map: an interactive map displaying nearby accessible places as map pins. Also, it includes search and filter controls.
- Forum: a community area where users browse and read posts within categories like Accessibility Tips and Product Reviews.
- Service: entry point for accessibility-related services such as (sign language interpreter, volunteer companion, education and jobs).
- Profile: personal profile and accessibility preferences for the logged-in user.

Prior to reaching the modules, the user interacts with the following interfaces: Sign Up and Login.

- The Sign-Up screen allows a new user to create an account using first name, last name, email/phone number and password, or by Google/Apple sign-in.
- The Login screen allows existing users to log in with their email/phone and password, or via Google/Apple.

Inline error messages appear below the corresponding fields when required information is missing.

Within the main modules:

- Home screen for quick access to featured places and recent forum posts and provides accessibility controls so users can adapt the interface to their needs.
- The Map screen helps users discover accessible locations using familiar map interactions such as search, zoom and tap on pins.
- The Forum screen is designed to show posts as cards with titles, short text, author, time, and engagement indicators, making that content easy to scan.
- The Service screen displays the service requests and education/ job opportunities as cards, with a clear label denoting status, like “Accepted”, “Completed” or “Accessible”.

- The Profile screen shows the user's account information and accessibility preferences, such as High Contrast Mode, Large Text, Screen Reader Support, Disability Type.

Overall, the interface is uniform across screens with clear typography and relying on icons and labels that correspond with real-world concepts allow immediate intuition of the app, both for a first-time and returning user: map pins, profile avatar, etc.

4.2 Interface Design Rules

Sanad's user interface follows common usability and accessibility principles to provide a predictable and user-friendly experience:

1. Consistency and standards:
 - The bottom navigation bar along with the general layout of header + content + bottom navigation bars are the same across all main screens.
 - Forms (Sign Up and Login) utilize consistent label placement, button styles and Inline validation messages.
2. Visibility of system status:
 - In the bottom navigation bar, the currently selected tab is always highlighted. Status labels, like "Accepted", "Completed" and "Accessible" appear on cards for services and opportunities to show status or type.
3. Match with the real world:
 - Icons and text labels reflect familiar objects and actions: map icon for Map, person icon for Profile, etc.
 - Terminology uses the language of accessibility services, such as "Sign Language Interpreter", "Volunteer Companion" and "Education & Jobs"
4. Recognition rather than recall:
 - Category chips (All, Education, Job, Workshop, Training) and tabs (Sign Language Interpreter, Volunteer Companion) allow users to select in a visual way rather than remembering commands.
 - Accessibility options are presented as visible toggles and buttons instead of hidden configuration codes.
5. Error prevention and simple error handling:
 - Login and Sign up forms validate their respective required fields before submission.
 - An inline error message of limited length appears directly beneath any given field when a required field has not been populated.
6. Aesthetic and minimalistic design:
 - The screens use a light background with a soft blue palette, enough whitespace and few primary actions, such as "Log In", "Sign Up", "Request Interpreter" and "View Details".
 - Grouping information in cards (places, posts, jobs and services) reduces cognitive load.
7. Accessibility support:

- The interface provides options like High Contrast, Text Size, Screen Reader Support and avoids clutter to support users with visual or cognitive impairments.
- Navigation icon and primary button touch targets are large enough for comfortable mobile interaction.

These rules guide future UI changes so that they stay consistent with the current design and with the accessibility goals of Sanad.

4.3 Screen Images

4.3.1 Home Interface

The Home screen displays:

- The Sanad title and a search bar at the top.
- A row of accessibility shortcut buttons: High Contrast, Text Size and Text-to-Speech.

Featured Places section displays the accessible places as cards, with name, accessibility description and a link for "View Details".

- A Recent Forum Posts section listing recent community posts as cards with Title, author and time.
- The bottom navigation bar with Home selected.

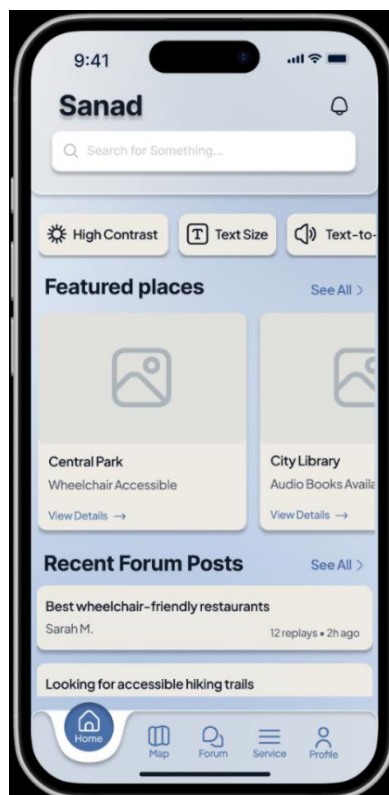


Figure 1- Home Interface: featured places, recent posts and accessibility shortcuts

4.3.2 Authentication Interfaces (Sign Up and Login)

Sign Up Screen:

- Input fields for First Name, Last Name, Email or Phone and Password.
- Primary "Sign Up" button.
- Optional sign-up with Google or Apple via dedicated buttons.
- A footer link directing existing users to the Login screen.

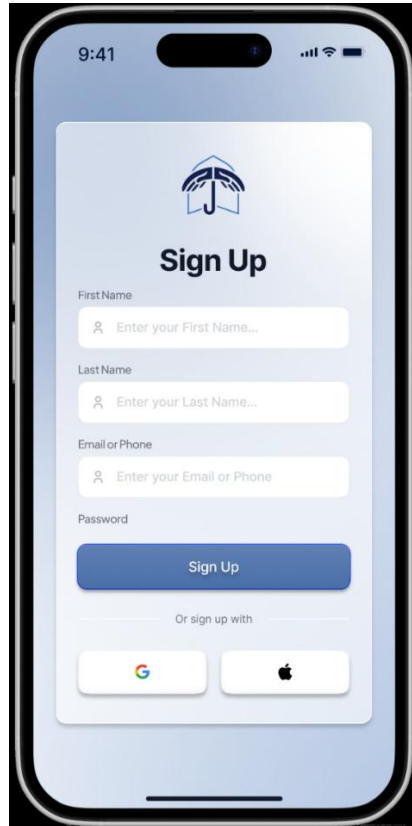


Figure 2- Sign Up Interface: creating a Sanad account

Login Screen:

- Input fields for Email or Phone and Password.
- A “Remember me” checkbox and a “Forgot Password?” link.
- A primary “Log In” button.
- Optional login with Google or Apple.
- When the required fields are empty, inline red error messages appear below them, such as "Email or phone is required" and "Password is required", etc.

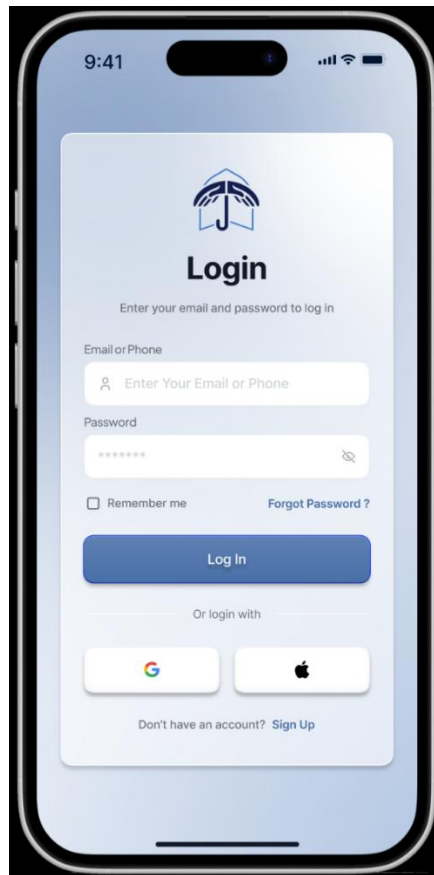


Figure 3- Login Interface: logging in with inline validation messages.

4.3.3 Map Interface

The Map screen shows:

- A top search bar for searching places.
- A filter icon next to the search bar.
- An interactive map view featuring multiple location pins showing places that are accessible.
- Floating circular buttons on the bottom right, such as zoom or options controls.
- The bottom navigation bar with Map highlighted.

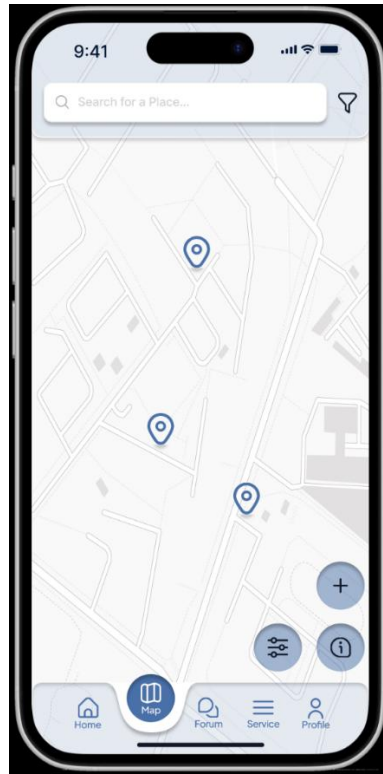


Figure 4- Map Interface: browsing accessible places on a map.

4.3.4 Service Interfaces

1. Service - Request Interpreter / Volunteer Companion:

- Tabs at the top for Sign Language Interpreter and Volunteer Companion.
- A large “Request Interpreter” button for creating a new interpreter request.
- A Request History section listing previous requests as cards with title, date, time, location and status such as (Accepted and Completed).
- Each card contains a “View Details” button.

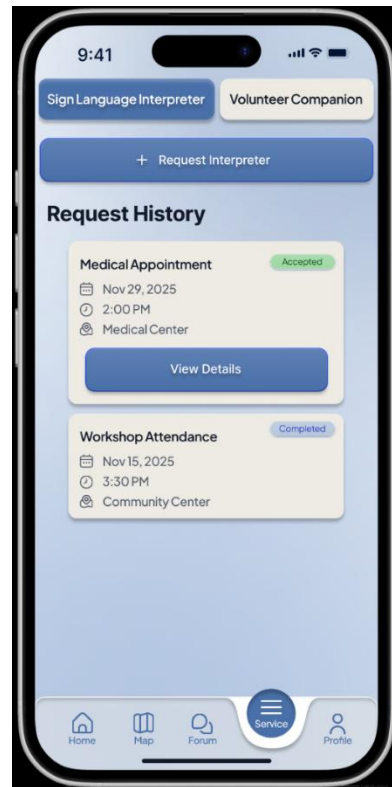


Figure 5- Service Interface: requesting services and viewing request history.

2. Service – Education & Jobs:

- Horizontal chips at the top to select the overall service category (Interpreter, Volunteer Companion, Education & Jobs, etc.).
- When Education & Jobs is active, a second row of chips (All, Education, Job, Workshop and Training) filters the list of opportunities.
- Each opportunity appears as a card with:
 - Category badge (Education, Job) and an “Accessible” badge.
 - Title, organization and short description.
 - Start or posted date and modality as (Remote).
 - A row of Accessibility Features.
 - A “View Details” button and a bookmark icon.

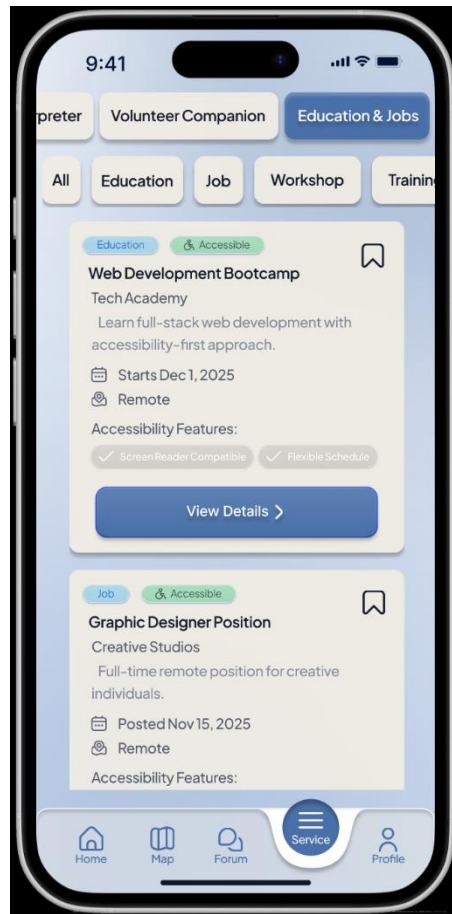


Figure 6- – Education & Jobs Interface: listing accessible educational and job opportunities.

4.3.5 Forum Interface

The Forum screen includes:

- Tabs for post categories, such as All, Accessibility Tips and Product Reviews.
- A vertical list of posts shown as cards with:
 - Title of post and a short preview.
 - Author name and role (Client, Family, etc.).
 - Time since posting.
 - Icons with numbers of comments and likes.
 - For some posts, a label such as “Approved by moderator”.

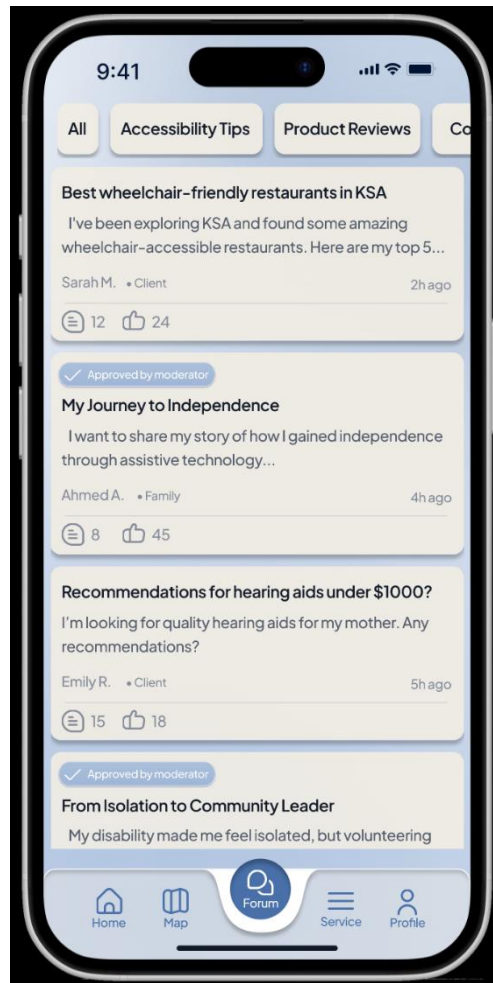


Figure 7- Forum Interface: browsing categorized community posts.

4.3.6 Profile Interface

The Profile screen displays:

- A profile header with user avatar, name, role (Client, Family, etc.), and membership date.
- An "Edit Profile" button.
- A Personal Information card with email, phone, and location.
- An Accessibility Preferences card with toggles for High Contrast Mode, Large Text and Screen Reader Support, plus a field for Disability Type.
- A section listing preferences or services related to accessibility as (an interpreter service with status "Completed").
- The bottom navigation bar with Profile highlighted.

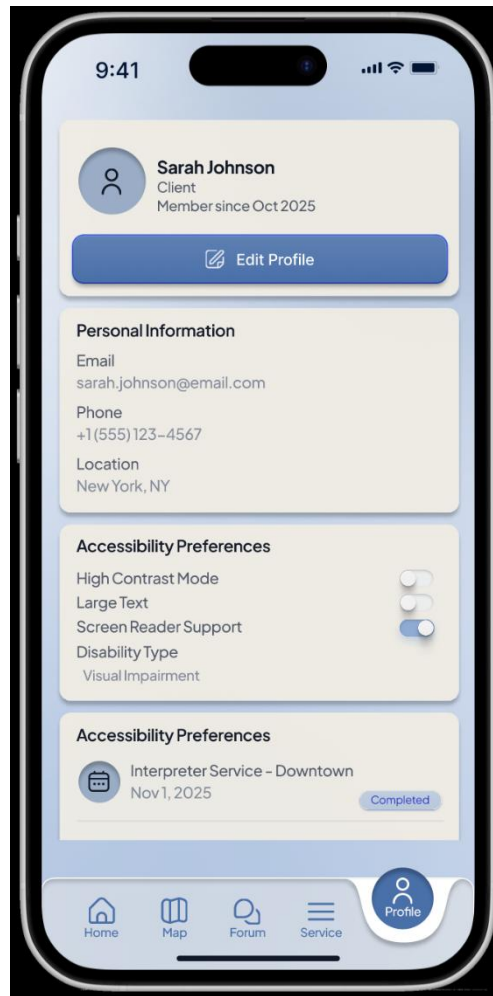


Figure 8- Profile Interface: personal information and accessibility preferences.

4.3.7 Admin Reports & Analytics Interface

For admin users, the Reports & Analytics screen shows:

- A header titled “Reports & Analytics”.
- A User Growth bar chart showing the number of users over days in a selected period (e.g., Nov 2025).
- An Activity Distribution donut/pie chart showing the distribution of different activities.
- An admin-specific bottom navigation bar with icons for different admin modules.



Figure 9-Admin Reports & Analytics Interface: visualizing user growth and activity distribution.

4.4 Screen Objects and Actions

Interface	Object	Type	Action
Home	Search bar	Text field	User enters a query; submitting searches for relevant places, services or content.
	High Contrast button	Button	Toggles high contrast visual mode for the app.
	Text Size button	Button	Increase base text size to improve readability.
	Text-to-Speech button	Button	Enables text-to-speech behavior where supported.
	Featured place card – “View Details”	Button / link	Opens detailed view of the selected place.

	Recent post card	Card	Opens the full forum post in the Forum module.
	Bottom navigation – Home	Navigation button	Displays Home screen (already active).
	Bottom navigation – Map		Switches to Map screen.
	Bottom navigation – Forum		Switches to Forum screen.
	Bottom navigation – Service		Switches to Service screen.
	Bottom nav – Profile		Switches to Profile screen.
Sign Up	First Name field	Text field	User enters first name for registration.
	Last Name field		User enters last name for registration.
	Email or Phone field		User enters email or phone; validated on submit.
	Password field	Password field	User enters password; validated on submit.
	Sign Up	Button	Submits registration; if valid, creates account and logs user in.
	Google button		Starts signing up using Google account.
	Apple button		Begin by registering with an Apple account.
	“Already have an account? Log In”	Link	Navigates to Login screen.
Login	Email or Phone field	Text field	User enters login identifier; if empty on submit, shows “Email or phone is required”.
	Password field	Password field	User enters password; if empty on submit, shows “Password is required”.
	Remember me	Checkbox	Marks login session to be remembered (per policy).
	Forgot Password?	Link	Goes to the password recovery process.
	Log In	Button	Successfully authenticates the user

			and takes them to the Home screen.
	Google button		Starts login using Google account.
	Apple button		Starts logging in using Apple account.
Map	Search bar	Text field	Filters locate places on the map by name.
	Filter icon	Button	Opens map filter options as (accessibility type).
	Map pin	Interactive icon	Tapping shows brief details or cards for that accessible place.
	Zoom / auxiliary buttons	Buttons	Adjust zoom level or open additional map options.
Service – Requests	Service tabs (Sign Language Interpreter, Volunteer Companion)	Tabs	Switch between service types and update the content area.
	Request Interpreter	Button	Starts or submits a new interpreter request.
	Request card	Card	Shows summary of a service request (title, date, time, location and status).
	Request card – “View Details”	Button / link	Opens detailed information about the selected request.
Service – Education & Jobs	Category chips (All, Education, Job, Workshop and Training)	Filter chips	Filter list of opportunities by selected category.
	Opportunity card	Card	Shows summary of an education/job opportunity.
	“View Details”	Button	Opens detailed information about the opportunity and its accessibility features.
	Bookmark icon	Toggle button	Adds or removes opportunity from the user’s saved list.
Forum	Category tabs (All, Accessibility Tips, Product Reviews, etc.)	Tabs	Filters display posts by category.
	Post card	Card	Opens full content of the selected forum post.

	Like icon / counter	Button	Registers a like for the post.
	Comment icon / counter		Opens the comments section for the post.
Profile	Edit Profile	Button	Opens profile editing screen.
	Personal Information fields (email, phone and location)	Display fields	Show user's stored personal information; editable from Edit Profile screen.
	High Contrast Mode toggle	Toggle switch	Enables or disables high-contrast mode and saves preference.
	Large Text toggle		Enables or disables large text mode and saves preference.
	Screen Reader Support toggle		Enables or disables additional support for screen readers.
	Disability Type field	Field / selector	Allows users to specify disability type used for personalization.
	Accessibility preference / service cards	Cards	Display related services or preferences may open details when tapped.
Admin Reports & Analytics	User Growth chart	Chart	Displays user growth metrics for a selected period.
	Activity Distribution chart		Displays distribution of user activities.
	Admin navigation icons	Navigation buttons	Switch between admin modules (overview, notifications and analytics).

Table 3- Screen Objects and Actions

4.5 Other Interfaces

This section covers additional interface behavior that is not fully described above, limited to what is currently visible in the designs.

4.5.1 Error Messages

Currently, the application performs inline validation on required fields on the Login or Sign-up screen using error messages:

- When the Email or Phone field value is empty and the user taps “Login or Sign up”, a red message appears below the field: “Email or phone is required”.
- When the Password field value is empty and the user taps “Login or Sign up”, a red message appears below the field: “Password is required”.
- When the Confirm Password field value is empty and the user taps “Login or Sign up”, a red message appears below the field: “Password Confirmation is required”.

These messages appear beside the respective input fields so that users can locate and fix the missing input conveniently.

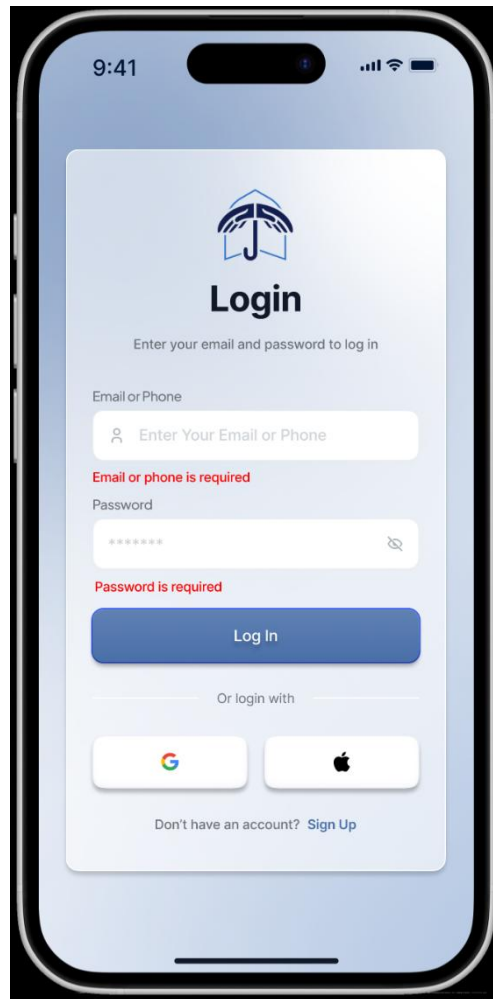


Figure 10- Login Error Interface.

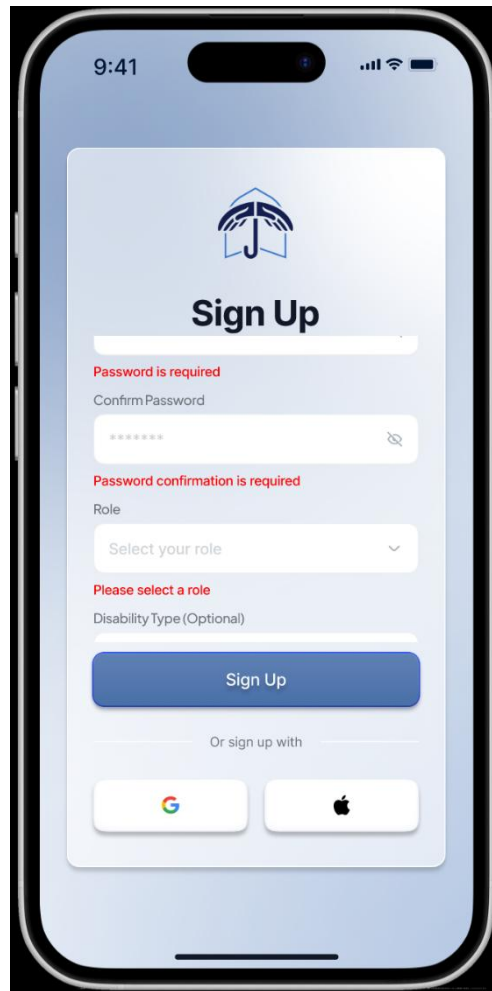


Figure 11-2 – Sign up Error Interface

4.5.2 Accessibility Settings Interfaces

Sanad provides explicit controls for accessibility configuration:

- On the Home screen, under the search bar are three shortcut buttons: High Contrast, Text Size and Text-to-Speech. These allow a user to change their accessibility settings quickly.
- Within the Profile screen, the Accessibility Preferences card contains toggle switches for High Contrast Mode, Large Text and Screen Reader Support. There is also a field called Disability Type, which is used for further personalization of the user's experience. These interfaces make accessibility options easy to find and modify, and once enabled, they are implemented throughout the rest of the application.

5. System Architecture

This section gives an overview of the high-level architecture of the Sanad Platform, describing how the system functions were split into subsystems, including how they all work together to enable accessibility with disabilities and able-bodied users together in community, service booking services, learning opportunities and access to assistive technologies. The decomposition method focused on modular design, scalability, and security while all meeting accessibility and privacy requirements. The subsections below describe the architectural design method, the overall structure of the system and the decomposition into internal subsystems.

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5.1 Architectural Design Approach

The ideal architectural design for the Sanad Platform is a layered, modular (multi tiered) design, which is very appropriate for web-based and mobile-enabled systems like Sanad. In this architecture, every major functionality and service (an accessibility map, a community forum, service and appointment booking, learning and employment, assistive technology library, etc.) is linked to its ideal layer. Thus, changes or future updates will be reflected primarily within the respective layer and modular portions of the system, not the entire system, and the design makes it easier to modify, extend and maintain over time.

Additionally, the layered technique diminishes dependencies between both modules and between the user interface and underlying infrastructure. This structure supports multi-platform implementations (web, iOS and Android) which allows the Sanad Platform to provide an experience that is consistent across devices while using the same core business logic as necessary. Further, the layered structure was selected by the development team, in part, because it supports multi-level security, which is a shared priority for Sanad, being that the platform will manage sensitive personal information, data about an accessibility concern, and a prospective history of locations and service use for individuals with disabilities. This also allows Sanad to ensure the system meets applicable security, privacy, and accessibility standards.

Sanad system layered architecture consists of four layers: Presentation (UI) Layer, Business (Application) Layer, Persistence Layer, and Data Layer.

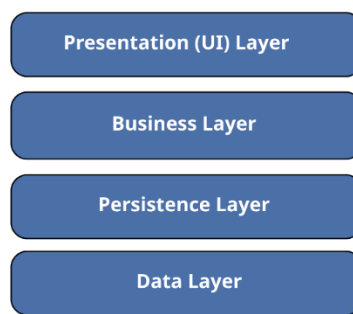


Figure 12-Architectural Design Approach

When compared to a layered approach, a non-layered architecture for the Sanad Platform would put the system at greater risk of external attacks and security and protection issues, particularly due to the sensitive nature of user profiles, accessibility data, and location information. Additionally, developing a portable and easily modifiable system will be much more difficult due to the challenges of one component of the system potentially causing unintended side effects to others. This approach will make maintenance, scale and multi platform support (web, iOS, and Android) much more difficult.

5.2 Architectural Design

The overall architecture of the Sanad Platform employs the layered architecture approach described in Figure (12).

The system is organized into four tiers: the Presentation (UI) Layer, Business Layer, Persistence Layer and Data Layer. Each layer encapsulates a specific set of responsibilities and interacts only with the adjacent layers. This modular architecture helps to separate user interaction, application logic, data access and data storage, which improves the maintainability, extensibility and security of the platform.

1. Presentation (UI) Layer:

The first layer manages all facets of engagement with end users (clients, volunteers, interpreters, and administrators). This layer displays information, accepts valid input, implements accessibility features. for example (high contrast or larger font sizes and structure for screen readers) and sends requests from users to the business layer. This includes: the web interface using (HTML, CSS and JavaScript/React) and the mobile interfaces (iOS/Android). Its main purpose of this layer is to ensure that all user groups can interact and have an engaging, accessible and simple experience.

2. Business Layer:

The core application functions and services are encapsulated in the second layer. This layer contains the business logic, responsible for numerous functionality that allows user registration and profile management, map search and rating, community/forum features or communication user features, service and appointment booking features, learning and employment features, assistive technology browsing and administrative and moderation functionality. The second layer also contains the workflows, validations and rules, events and exceptions, and also acts as a bridge to the Persistence Layer. for example (when the second layer is resolving which data will have to be read, created or modified).

3. Persistence Layer:

The third layer handles the access and management of data in a controlled manner. It implements repositories/data access components that handle queries from the business layer and thereby enforce role based access restrictions according to user permissions (admin versus simple user). It hides the details of the database and any external APIs from the business logic. It may also include: but is not limited to, functionalities such as object relational mapping (ORM), caching and integration with external services such as map APIs, video-call APIs and job/training feeds. The data layer is only directly exposed to the third layer.

4. Data Layer:

The lowest layer of the architecture is the Data Layer. It presents the relational database for example, MySQL. that stores, retrieves, updates and deletes all persistent data of the system, such as user accounts, locations, bookings, forum posts, reviews, courses, jobs and assistive technology items. This layer covers all media and any external services, such as image files, documents and connections to external servers and APIs (map services, notification/email services, or payment/authentication

providers). This layer is responsible for secure, permanent storage of the actual data of the Sanad Platform and making a backup of the data as required and data recovery if needed.

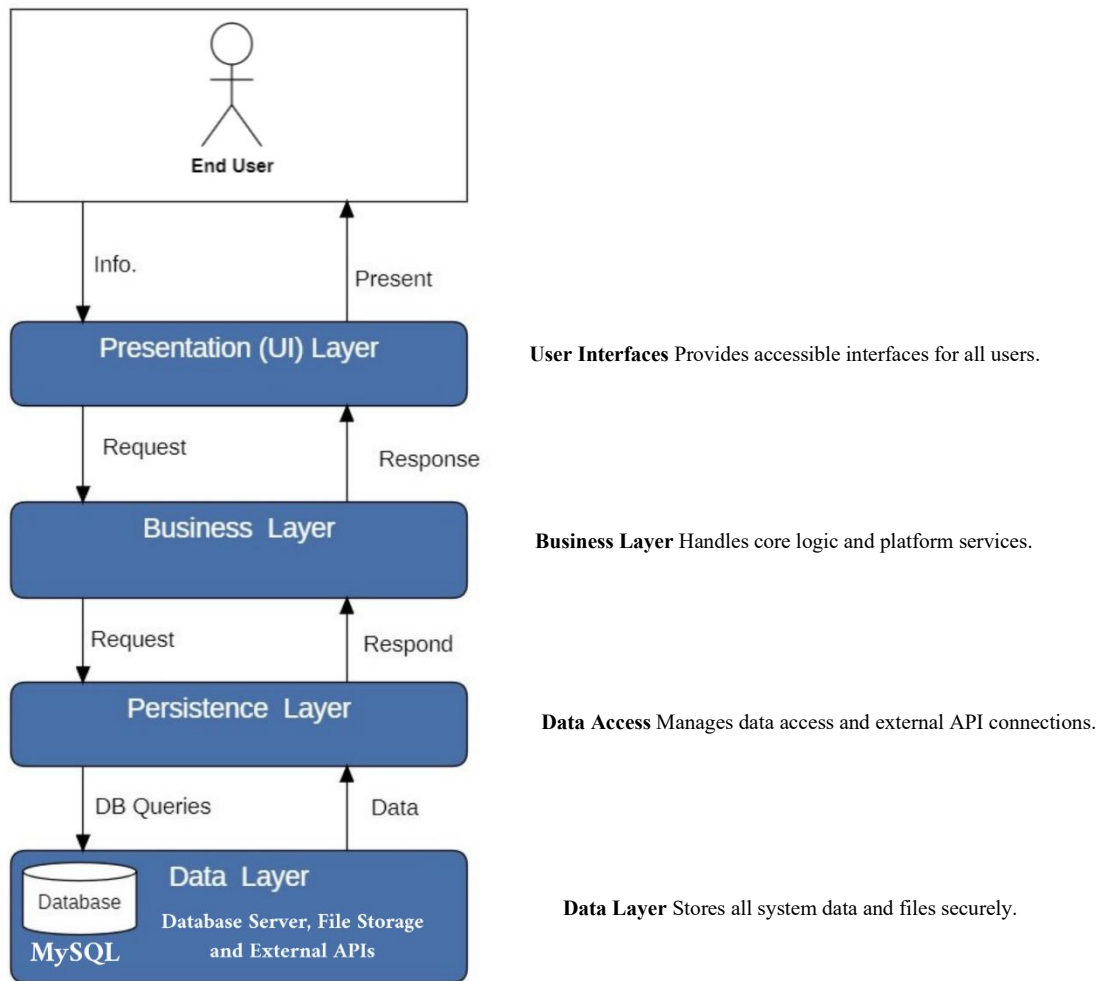


Figure 13-Architectural Design

5.3 Subsystem Architecture

This section explains the subsystem organization of the Sanad platform through a functional, data flow based perspective. The system is decomposed into multiple logical subsystems, each of which serves as a home for a grouping of services: access and profiles, accessibility map, community and support, service booking, learning and employment resources, notifications and administration.

They are presented to clarify the movement of data between participants external to the system and internal processes of Sanad. Each subsystem is described, accompanied by a data flow diagram (DFD). They start with a Context DFD that presents Sanad as a single process interacting with external participants. Then, each subsystem is presented with a Level 0 DFD that describes The subsystem decomposition, in terms of the primary processes, data stores and data flows, of the system it comprises.

5.3.1 General View of the System

A context DFD is used to represent the whole Sanad system in a simple manner. It is used to illustrate the system's boundaries, limitations and its relationship with its environment. Figure (12) shows how data is interchanged between the external actors and the Sanad platform.

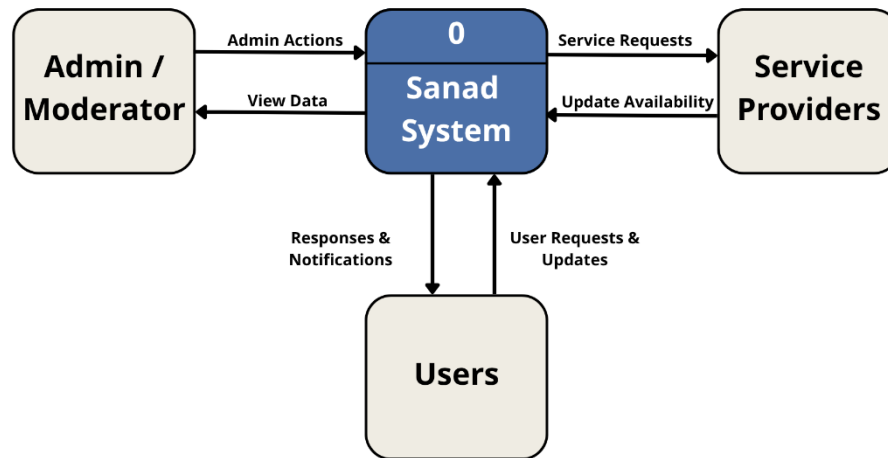


Figure 14-General View of the System

5.3.2 User Management & Authentication Subsystem

A level 0 DFD is used here to describe the main data flows of the User Management & Authentication Subsystem process in the Sanad platform. Figure (15)

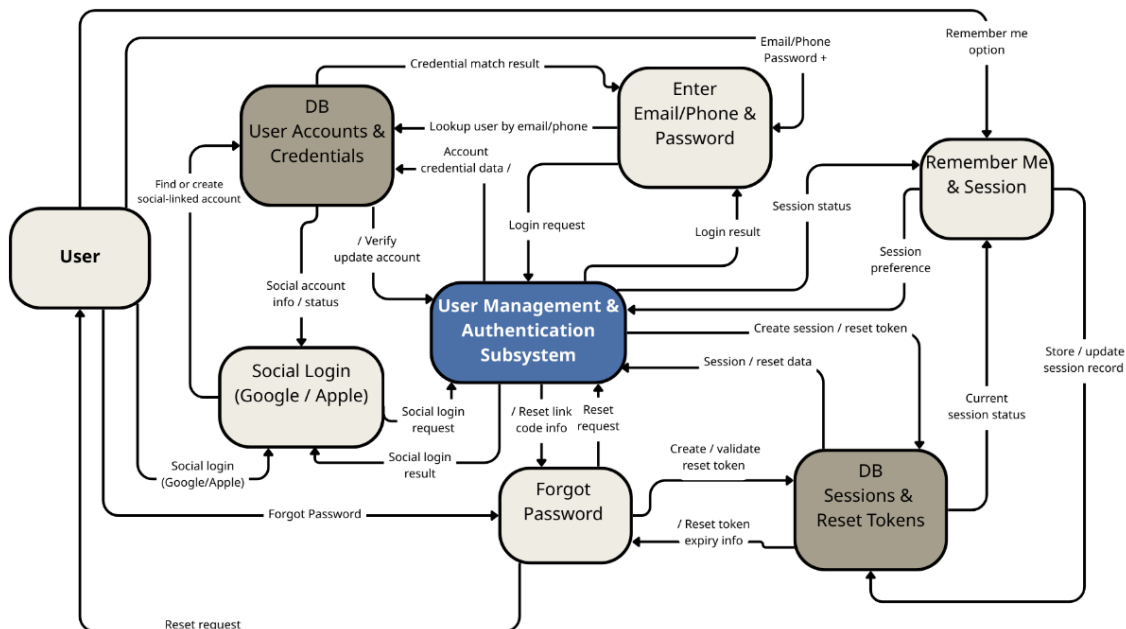


Figure 15-User Management & Authentication Subsystem

5.3.3 Sign Up & User Profile Initialization

A level 0 DFD is used to describe the data flow for the Sign Up & User Profile Initialization of the Sanad platform. Figure (16)

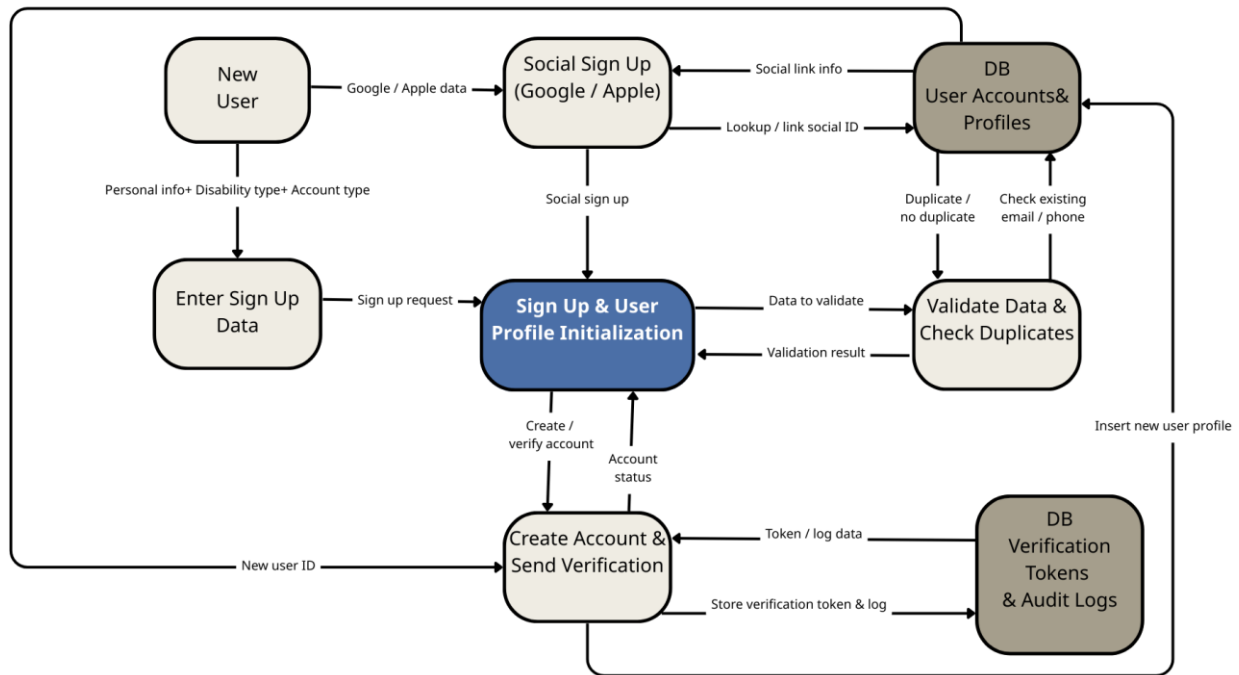


Figure 16-Sign Up & User Profile Initialization

5.3.4 Home Dashboard & Personalization

A level 0 DFD is used to describe the data flow for the Home Dashboard & Personalization of the Sanad platform. Figure (16)

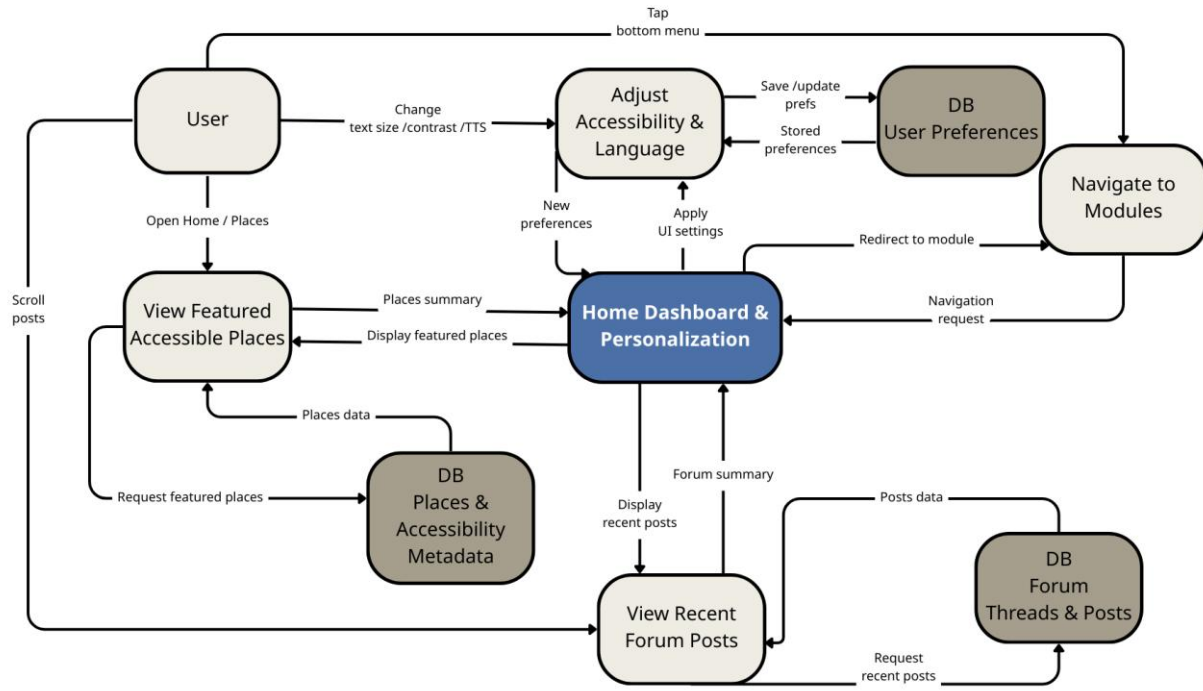


Figure 17-Home Dashboard & Personalization

5.3.5 Map & Accessible Places Subsystem

A level 0 DFD is used to describe the data flow for the Map & Accessible Places Subsystem of the Sanad platform. Figure (18)

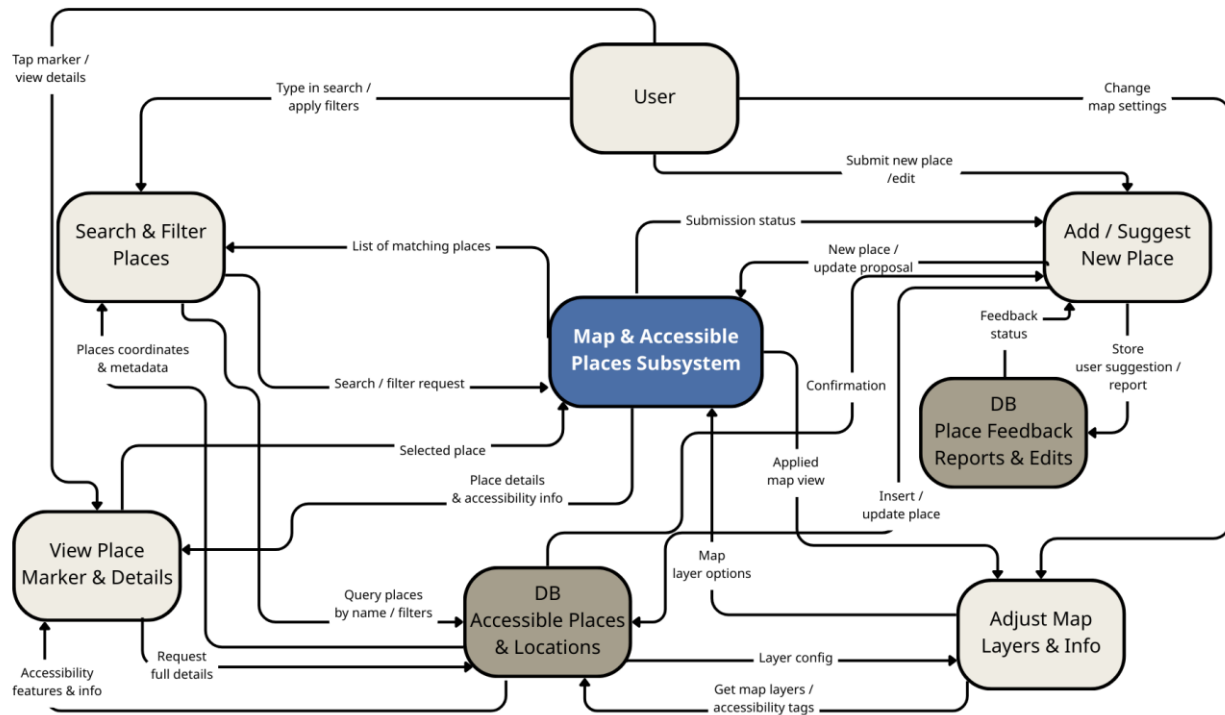


Figure 18-Map & Accessible Places Subsystem

5.3.6 Community & Forum Subsystem

A level 0 DFD is used to describe the data flow for the Community & Forum Subsystem of the Sanad platform. Figure (19)

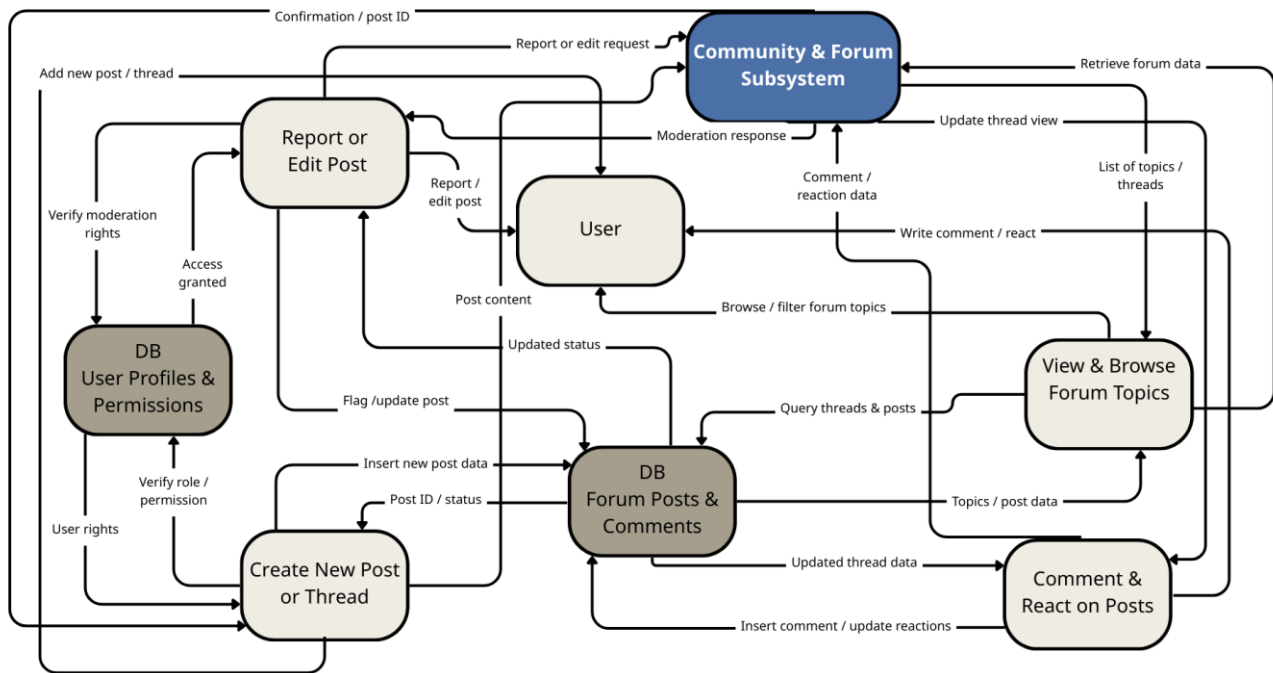


Figure 19-Community & Forum Subsystem

5.3.7 Service & Appointment Booking Subsystem

A level 0 DFD is used to describe the data flow for the Service & Appointment Booking Subsystem of the Sanad platform. Figure (20)

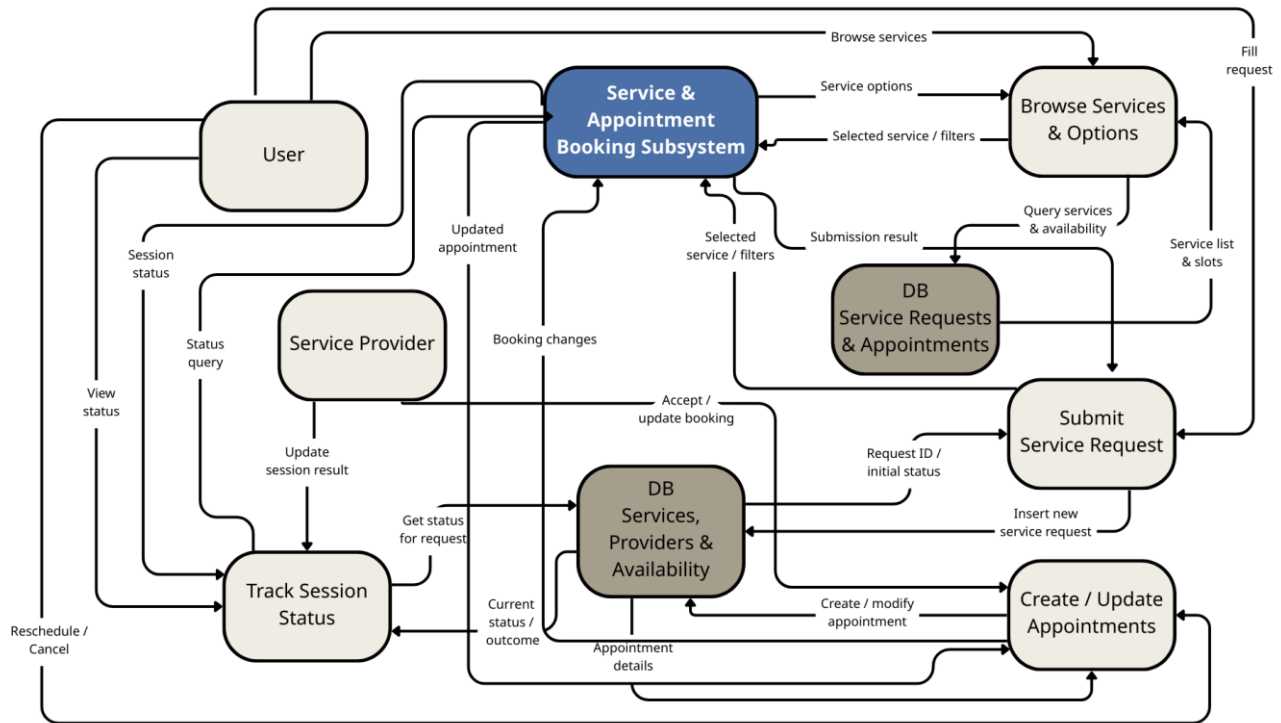


Figure 20- Service & Appointment Booking Subsystem

5.3.8 Profile & Accessibility Preferences

A level 0 DFD is used to describe the data flow for the Profile & Accessibility Preferences Subsystem of the Sanad platform. Figure (21)

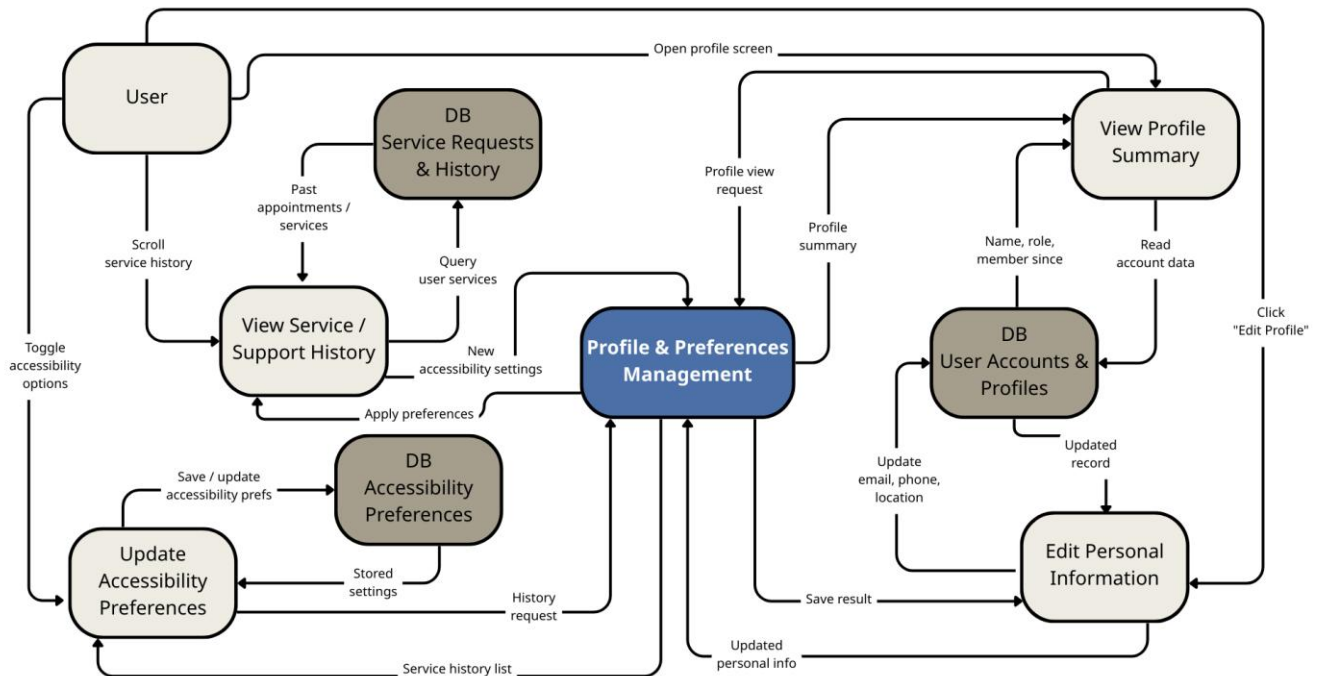


Figure 21- Profile & Accessibility Preferences

6. Data Design

6.1 Data Description

Entity	Field	Type	PK	FK	Constraints
User	UserID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	FirstName	VARCHAR(10)	—	—	NOT NULL
	LastName	VARCHAR(10)	—	—	NOT NULL
	Email	VARCHAR(60)	—	—	NOT NULL, UNIQUE
	Phone	VARCHAR(20)	—	—	NOT NULL
	Password	VARCHAR(20)	—	—	NOT NULL
	Role	ENUM('User','Volunteer','Interpreter','Admin')	—	—	NOT NULL
	AccessibilityPrefs	JSON	—	—	NULL
	CreatedAt	DATETIME	—	—	NOT NULL

Table 4- Data Description "User "

Entity	Field	Type	PK	FK	Constraints
Place	PlaceID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	Name	VARCHAR(80)	—	—	NOT NULL
	Address	VARCHAR(200)	—	—	NOT NULL
	Latitude	DOUBLE	—	—	NOT NULL
	Longitude	DOUBLE	—	—	NOT NULL
	Category	VARCHAR(40)	—	—	NOT NULL
	AccessibilityFeatures	JSON	—	—	NULL
	AddedByUserID	INT(10)	—	✓	FK → User(UserID)

Table 5-Data Description "Place "

Entity	Field	Type	PK	FK	Constraints
Review	ReviewID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	UserID	INT(10)	—	✓	FK → User(UserID)
	PlaceID	INT(10)	—	✓	FK → Place(PlaceID)
	Rating	INT(1)	—	—	CHECK (Rating BETWEEN 1 AND 5)
	ReviewText	LONGTEXT	—	—	NULL
	CreatedAt	DATETIME	—	—	NOT NULL

Table 6 - Data Description "Review "

Entity	Field	Type	PK	FK	Constraints
ForumPost	PostID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	UserID	INT(10)	—	✓	FK → User(UserID)
	Category	VARCHAR(40)	—	—	NOT NULL
	Title	VARCHAR(200)	—	—	NOT NULL
	Content	LONGTEXT	—	—	NOT NULL
	CreatedAt	DATETIME	—	—	NOT NULL
	Status	ENUM('Active','Flagged','Archived')	—	—	NOT NULL

Table 7- Data Description "ForumPost "

Entity	Field	Type	PK	FK	Constraints
Comment	CommentID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	PostID	INT(10)	—	✓	FK → ForumPost(PostID)
	UserID	INT(10)	—	✓	FK → User(UserID)
	Content	LONGTEXT	—	—	NOT NULL
	CreatedAt	DATETIME	—	—	NOT NULL

Table 8- Data Description "Comment "

Entity	Field	Type	PK	FK	Constraints
Service	ServiceID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	ServiceType	ENUM('Interpreter','VolunteerEscort')	—	—	NOT NULL
	Description	LONGTEXT	—	—	NULL

Table 9- Data Description "Service "

Entity	Field	Type	PK	FK	Constraints
Booking	BookingID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	UserID	INT(10)	—	✓	FK → User(UserID)
	ServiceID	INT(10)	—	✓	FK → Service(ServiceID)
	ProviderID	INT(10)	—	✓	FK → User(UserID)
	DateTime	DATETIME	—	—	NOT NULL
	Status	ENUM('Pending','Confirmed','Completed','Cancelled')	—	—	NOT NULL

Table 10- Data Description "Booking "

Entity	Field	Type	PK	FK	Constraints
Course	CourseID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	Title	VARCHAR(120)	—	—	NOT NULL
	Description	LONGTEXT	—	—	NULL
	Provider	VARCHAR(80)	—	—	NOT NULL
	DurationHours	INT(5)	—	—	NULL

Table 11- Data Description "Course "

Entity	Field	Type	PK	FK	Constraints
Enrollment	EnrollmentID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	UserID	INT(10)	—	✓	FK → User(UserID)
	CourseID	INT(10)	—	✓	FK → Course(CourseID)

	Progress	INT(3)	—	—	CHECK (Progress BETWEEN 0 AND 100)
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Table 12- Data Description " Enrollment "

Entity	Field	Type	PK	FK	Constraints
AssistiveProduct	ProductID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	Name	VARCHAR(100)	—	—	NOT NULL
	Category	VARCHAR(50)	—	—	NOT NULL
	Description	LONGTEXT	—	—	NULL
	Price	DOUBLE	—	—	NULL

Table 13- Data Description "AssistiveProduct "

Entity	Field	Type	PK	FK	Constraints
SupportRequest	RequestID	INT(10)	✓	—	NOT NULL, AUTO_INCREMENT
	UserID	INT(10)	—	✓	FK → User(UserID)
	ProductID	INT(10)	—	✓	FK → AssistiveProduct (ProductID)
	Description	LONGTEXT	—	—	NULL
	Status	ENUM('Pending','Approved','Rejected')	—	—	NOT NULL

Table 14- Data Description "SupportRequest "

6.2 Data Dictionary

Entity	Fields	Description
User	UserID	Unique identification number for each user, must be unique.
	FirstName	The user's first name, cannot be empty.
	LastName	The user's last name, cannot be empty.
	Email	The user's email, must be unique and not empty.
	Phone	The user's phone number, must not be empty.
	Password	The encrypted password of the user.
	Role	Defines whether the user is a regular user, volunteer, interpreter, or admin.

	AccessibilityPrefs	The user's accessibility preferences (contrast, text size, etc.).
	CreatedAt	The creation date of the user account.

Table 15- Data Dictionary "User "

Entity	Fields	Description
Place	PlaceID	Unique identification number for each accessible place.
	Name	The name of the accessible place.
	Address	The full address of the place.
	Latitude	The latitude coordinate of the place.
	Longitude	The longitude coordinate of the place.
	Category	The category of the place (restaurant, café, hospital, etc.).
	AccessibilityFeatures	A list of accessibility features (ramp, elevator, braille signs...).
	AddedByUserID	The ID of the user who submitted the place.
	Status	The approval status (Pending / Approved / Rejected).

Table 16- Data Dictionary "Place "

Entity	Fields	Description
Review	ReviewID	Unique identification number for each review.
	UserID	The ID of the user who wrote the review.
	PlaceID	The ID of the place being reviewed.
	Rating	The numeric rating (1–5).
	ReviewText	The written feedback.
	CreatedAt	The creation date of the review.

Table 17- Data Dictionary "Review "

Entity	Fields	Description
ForumPost	PostID	Unique identification number for each forum post.
	UserID	The ID of the user who created the post.
	Category	The forum category (tips, stories, reviews...).
	Title	The title of the forum post.
	Content	The body text of the forum post.
	CreatedAt	The date the post was created.
	Status	Indicates whether the post is active, flagged, or archived.

Table 18- Data Dictionary "ForumPost "

Entity	Fields	Description
Comment	CommentID	Unique identification number for each comment.
	PostID	The ID of the forum post the comment belongs to.
	UserID	The ID of the user who wrote the comment.
	Content	The text content of the comment.
	CreatedAt	The date the comment was created.

Table 19- Data Dictionary "Comment "

Entity	Fields	Description
Service	ServiceID	Unique identification number for the service.
	ServiceType	Defines if the service is "Interpreter" or "Volunteer Escort".
	Description	Detailed description of the service (optional).

Table 20- Data Dictionary "Service "

Entity	Fields	Description
Booking	BookingID	Unique ID for each service booking.
	UserID	The ID of the user making the booking.
	ServiceID	The ID of the requested service.
	ProviderID	The ID of the volunteer/interpreter providing the service.
	DateTime	The scheduled date and time.
	Status	Booking status (Pending, Confirmed, Completed, Cancelled).

Table 21- Data Dictionary "Booking "

Entity	Fields	Description
Course	CourseID	Unique identification number for each course.
	Title	The title of the course.
	Description	A detailed description of the course.
	Provider	The organization or entity offering the course.
	DurationHours	Total duration of the course.

Table 22- Data Dictionary "Course "

Entity	Fields	Description
Enrollment	EnrollmentID	Unique ID for each course enrollment record.
	UserID	The user who is enrolled.
	CourseID	The course the user is enrolled in.
	Progress	User's progress in the course (0–100%).

Table 23- Data Dictionary "Enrollment"

Entity	Fields	Description
AssistiveProduct	ProductID	Unique ID for each assistive product.
	Name	The name of the product.
	Category	The product category (visual aid, mobility aid ...).
	Description	A description of the product.
	Price	The price of the product (optional).

Table 24- Data Dictionary "AssistiveProduct "

Entity	Fields	Description
SupportRequest	RequestID	Unique ID for each support request.
	UserID	The ID of the user requesting support.
	ProductID	The product related to the support request.
	Description	The reason or details of the support request.
	Status	Support request status (Pending, Approved, Rejected).

Table 25- Data Dictionary "SupportRequest "

6.3 Database Description

Sanad Platform and the relationships between them. It presents how users interact with places, accessibility features, reviews, service requests, and appointments. The diagram reflects all logical data connections required to support the system functionalities described in Sections 6.1 and 6.2.

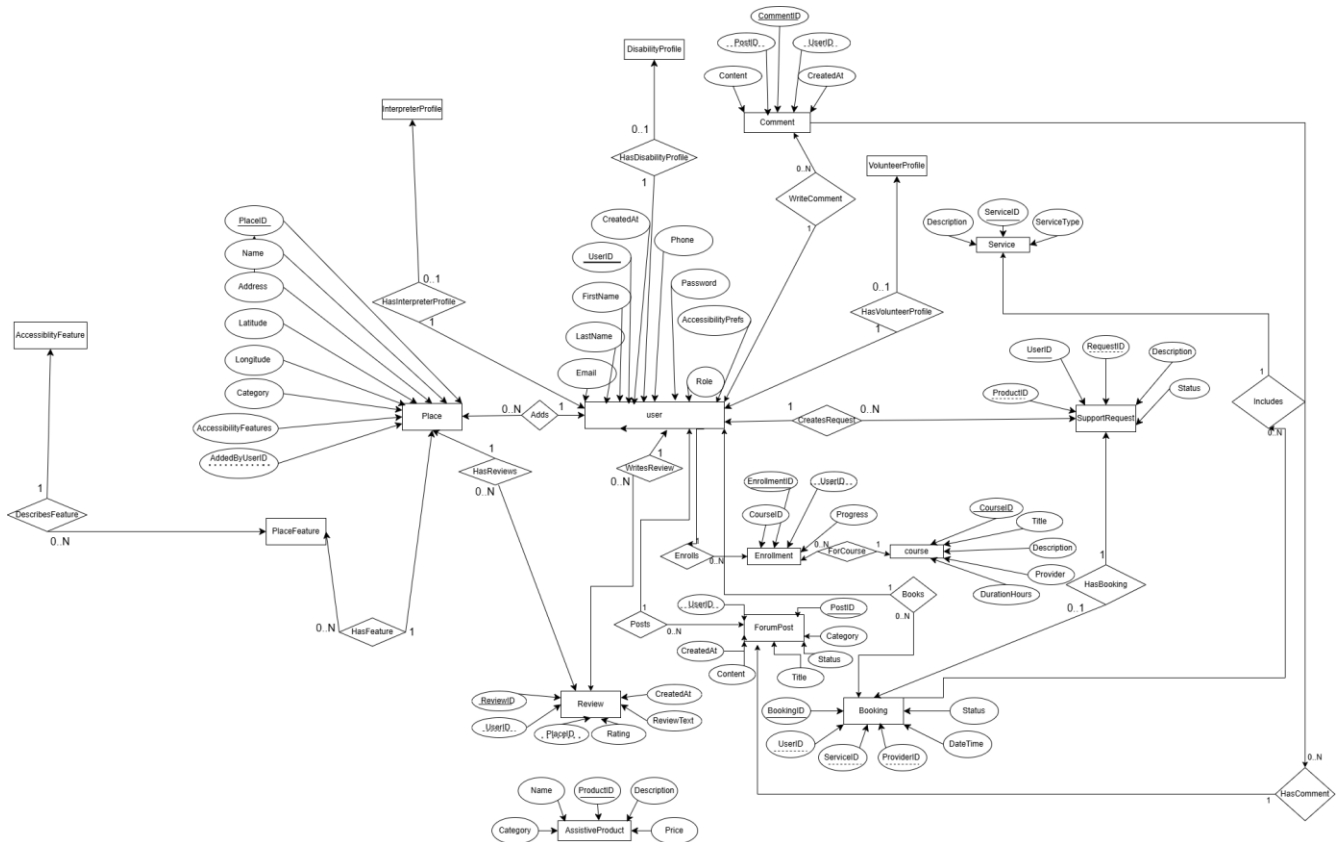


Figure 22–Database Description

7. Component Design

In this section, pseudocode of Sanad's major functionalities is provided to give an informal description of the system's main functions. The pseudocode is expressed in natural language and is based on the behavior visible in the user interfaces.

7.1 Common Functions

7.1.1 Sign Up

SignUp(){

 Input FirstName, LastName, EmailOrPhone, Password, ConfirmPassword, Role

 If "Sign Up" button is clicked then

 If Password is empty then

 Display "Password is required"

 EndIf

 If ConfirmPassword is empty then

 Display "Password confirmation is required"

 EndIf

 If Role is not selected then

 Display "Please select a role"

 EndIf

 If all required fields are filled AND Password = ConfirmPassword AND Role is selected then

 Create new user account with the selected Role

 Save user information in the database

Redirect user to the main Home interface

Else

Stay on the Sign Up interface and allow the user to correct inputs

EndIf

EndIf

}

7.1.2 Sign In

SignIn(){

Input EmailOrPhone, Password

If “Log In” button is clicked then

If EmailOrPhone is empty then

Display “Email or phone is required”

EndIf

If Password is empty then

Display “Password is required”

EndIf

If EmailOrPhone is not empty AND Password is not empty then

Verify credentials with the database

If credentials are correct then

If user Role = “Admin” then

Redirect to Admin Reports & Analytics interface

Else

Redirect to main Home interface

EndIf

Else

Display generic message “Invalid email/phone or password”

Allow user to re-enter EmailOrPhone and Password

EndIf

EndIf

EndIf

}

7.2 Client Functions

7.2.1 Manage Home

Manage_Home(){

When Home interface is loaded then

Display search bar, accessibility shortcut buttons, featured places, and recent forum posts

If user enters text in search bar and submits then

Search for matching places, services, or posts

Display filtered results in the content area

EndIf

If “High Contrast” button is toggled then

Enable or disable high-contrast theme across the app

EndIf

If “Text Size” button is toggled then

Increase or reset base font size for text

EndIf

If “Text-to-Speech” button is toggled then

Enable or disable text-to-speech support where available

EndIf

If user taps a Featured Place card or its “View Details” link then

Open place details interface for that location

EndIf

If user taps a Recent Forum Post card then

Open full forum post in Forum module

EndIf

}

7.2.2 View Accessible Map

View_Map(){

When Map interface is loaded then

Retrieve nearby accessible places from database or service

Display map with pins for each accessible place

If user enters text in map search bar and submits then

Filter places by name

Update pins shown on the map

EndIf

If user clicks Filter icon then

Display filter options (such as accessibility type, category, distance)

Apply selected filters

Refresh pins on the map

EndIf

If user taps a map pin then

Show brief place information card

Allow user to open full details for that place

EndIf

}

7.2.3 Request Interpreter / Volunteer Companion

Manage_Services_Request(){

When Service tab is opened then

Display tabs: “Sign Language Interpreter”, “Volunteer Companion”, “Education & Jobs”

If “Sign Language Interpreter” tab is active then

Display “Request Interpreter” button

Display list of previous interpreter requests with title, date, time, location, and status

EndIf

If “Volunteer Companion” tab is active then

Display list of companion requests (if defined) similarly

EndIf

If user clicks “Request Interpreter” then

Open interpreter request form or create request with selected data

Validate required fields (such as date, time, location)

If request data is valid then

Save new interpreter request in database

Update Request History list

Else

Show appropriate validation messages on the form

EndIf

EndIf

If user taps a request card or its “View Details” button then

Open detailed view for the selected request (including status and full info)

EndIf

}

7.2.4 Browse Education & Jobs

Manage_Education_Jobs(){

When “Education & Jobs” sub-tab is selected in Service then

Display filter chips: All, Education, Job, Workshop, Training

Retrieve matching opportunities from database

Display opportunities as cards with title, organization, brief description, dates, modality, and accessibility features

If user selects a filter chip (such as Job) then

Filter opportunities by selected type

Refresh list of cards

EndIf

If user taps an opportunity card or its “View Details” button then

Open detailed view with full description, requirements, and accessibility details

EndIf

If user taps bookmark icon on an opportunity then

If opportunity is not yet bookmarked for this user then

Add opportunity to user’s saved list

Else

Remove opportunity from user’s saved list

EndIf

Update bookmark icon state

EndIf

}

7.2.5 Browse Forum

Manage_Forum(){

 When Forum tab is opened then

 Display category tabs: All, Accessibility Tips, Product Reviews, etc.

 Load recent posts from database

 Display posts as cards with title, snippet, author, time, likes, and comments count

 If user selects a category tab then

 Filter posts by selected category

 Refresh list

 EndIf

 If user taps a post card then

 Open full post details, including comments list

 EndIf

 If user clicks Like icon on a post then

 Toggle like status for that post for the current user

 Update like counter on the card

 EndIf

 If user enters a comment text and submits then

 Save comment to database

 Refresh comments displayed under the post

 EndIf

}

7.2.6 Manage Profile and Accessibility Preferences

Manage_Profile(){

When Profile tab is opened then

Retrieve user account information and accessibility preferences

Display profile header, personal information card, and accessibility preferences card

If “Edit Profile” button is clicked then

Open profile editing interface

Allow user to modify editable fields (such as email, phone, location)

If user clicks “Save” then

Validate new data

If data is valid then

Save updated information in database

Return to Profile interface and show updated data

Else

Display relevant validation messages

EndIf

EndIf

EndIf

If “High Contrast Mode” toggle is changed then

Update user preference value

Apply or remove high-contrast theme globally

EndIf

If “Large Text” toggle is changed then

Update user preference value

Increase or reset base font size in the app

EndIf

If “Screen Reader Support” toggle is changed then

Update user preference value

Enable or disable additional accessibility labels and hints

EndIf

If user selects or changes “Disability Type” then

Save selected disability type in user profile

Use it to personalize recommendations and default accessibility options

EndIf

}

7.3 Admin Functions

7.3.1 View Reports and Analytics

View_Admin_Reports(){

When an Admin user logs in successfully then

Redirect to Reports & Analytics interface

When Reports & Analytics interface is loaded then

Retrieve aggregated usage statistics for the default period

Display User Growth bar chart

Display Activity Distribution donut chart

If admin changes reporting period or applies a filter then

Recalculate statistics for the selected criteria

Update bar chart and donut chart with new data

EndIf

(Optional) If admin taps a specific bar or segment in a chart then

Display additional details for that day or activity type

EndIf

}

7.3.2 Moderate Forum Content

Moderate_Forum(){

When moderator opens forum moderation interface then

Retrieve all posts whose status is “Pending”

For each pending post do

Display post content, author, and submission time

If moderator clicks “Approve” then

Set post status to “Approved”

Make post visible in main Forum list

Attach “Approved by moderator” label to post card

EndIf

If moderator clicks “Reject” then

Set post status to “Rejected”

Optionally notify post author with rejection reason

EndIf

EndFor

}

7.3.3 Manage User Accounts

Manage_Users(){

When admin opens user management interface then

 Retrieve list of users with their roles and account status

If admin enters search text (name or email) then

 Filter users list based on search text

EndIf

If admin selects a user from list then

 Display full user profile and summary of activity

EndIf

If admin changes user Role (such as Client to Admin) or deactivates account then

 Save changes in database

 Apply new permissions the next time user signs in

EndIf

}

8. Detailed System Design

The following section provides a comprehensive description of the components and modules of the Sanad Platform system. Various component characteristics, such as classification, definition, responsibilities, limitations, configuration, uses/interactions, processing, resources, and exports, will be discussed in detail. Sequence diagrams will be provided to illustrate the relationships and interactions between components. Additionally, flowcharts will be used to demonstrate the internal structure of each component and provide a detailed view of the information flow.

8.1 Classification

In the following sub section, each component's type, objectives, and responsibilities shall be described with the help of the Table:

Component	Classification	Definition
Login Interface		
Login Screen	User Interface Component - Module	The visual interface used by users to enter their credentials.
Authentication Controller	Module - Controller	Handles login requests, validations, and forwards credentials to authentication service.
Authentication Service	System Service	Verifies user identity using email - phone and password, communicates with database.
User Credentials Form	UI Subcomponent - Form	Input fields for email - phone and password.
Password Visibility Toggle	UI Function	Allows users to show - hide password characters.
Remember Me	UI Function - Local Storage Handler	Saves session preferences on user device if enabled.
Forgot Password	Function - Navigation Link	Redirects the user to password recovery workflow.
Social Login (Google)	External Auth Integration Component	Allows login using Google OAuth services.
Social Login (Apple)	External Auth Integration Component	Allows login using Apple ID authentication.
Input Validation Module	Module - Function	Validates format of email - phone and password before sending request.
Session Manager	System Component	Creates and stores authenticated sessions once login succeeds.
Error Handler	Component - Function	Displays login errors such as wrong password or invalid input.

Sign Up Interface		
SignUpScreen	UI Screen - Module	A user-facing interface that allows new users to create an account by entering their personal information. It serves as the entry point for system registration.
InputField FullName	UI Element - Class	Captures the user's full name as part of the registration form.
InputField Email	UI Element - Class	Accepts the user's email address and performs basic format validation.
InputField Password	UI Element - Class	Secure password input field with masking for privacy.
InputField ConfirmPassword	UI Element - Class	Ensures the user re-types the same password for verification.
Button SignUpButton	UI Control - Function Trigger	Submits the registration form and triggers the account-creation process.
Button LoginRedirectButton	Navigation UI Control	Navigates the user to the Login screen if they already have an account.
ValidationService	Service - Function	Validates the user inputs (email format, password match, empty fields).
UserAccountService	Backend Service - Subsystem	Handles account creation, communicates with database, and stores new user records.
Database UsersTable	Data Storage Component	Stores registered users' credentials and related profile information.
Home Screen Interface		
HomeScreen	UI Screen - Module	The main landing page of the system, showing quick access to the core features.
Button AccessibilityMap	Navigation UI Control	Redirects users to the accessible-map interface.
Button ServicesSection	Navigation UI Control	Opens the services and booking interface.
Button ReportsAnalytics	Navigation UI Control	Redirects users to the reports & analytics section.
Button ForumSection	Navigation UI Control	Opens the community support and forum screen.
Button ProfileSection	Navigation UI Control	Opens the user profile and settings screen.
HeaderComponent	UI Element	Displays the app title - logo.
QuickActionsPanel	UI Container	Shows shortcuts to commonly used actions.
Reports & Analytics Screen Interface		

ReportsAnalyticsScreen	UI Screen - Module	Displays user reports, statistics, and analytics visualizations.
ReportCard	UI Component - Class	Shows summarized report information in a card format.
AnalyticsChart	UI Visualization Component	Graphical representation of user or system data.
FilterPanel	UI Element	Allows users to filter report data.
Button ViewDetailedReport	UI Control - Function Trigger	Opens a full report with detailed information.
DataAnalyticsService	Backend Service	Processes and aggregates data for analytics.
ReportsDatabase	Data Storage Component	Stores generated reports and analytics history.
Map Screen Interface		
MapScreen	UI Screen - Module	Displays an interactive map showing accessible places for people with disabilities.
MapWidget	UI Component	Renders the map interface with zoom and navigation.
MarkerComponent	UI Element	Represents accessible locations on the map.
SearchBar	UI Element	Allows users to search for places or services.
FilterButton	UI Control	Applies accessibility- based filters to map results.
LocationService	Backend Service - API	Retrieves map data and place accessibility information.
GeoDatabase	Data Storage Component	Stores geo-location and accessibility metadata.
Forum Screen Interface		
ForumScreen	UI Screen - Module	Provides community support, discussions, and user-generated posts.
PostListComponent	UI Component	Displays forum posts.
PostItem	UI Class	Represents a single forum post entry.
CreatePostButton	UI Control	Opens the new –post creation form.
ReplyComponent	UI Sub-Component	Displays replies under each post.
ForumService	Backend Service	Handles posting, retrieving, and updating forum content.
ForumDatabase	Data Storage	Stores forum posts, replies, and user interactions.
Services Screen Interface		
ServicesScreen	UI Screen - Module	Shows available services such as booking interpreters or requesting assistance.
ServiceCard	UI Component	Displays service details with icon and description.
BookingButton	UI Control	Opens the booking interface for the selected

		service.
ServiceCategoryFilter	UI Element	Filters services by category or type.
ServiceBookingService	Backend Service	Manages booking requests and validations.
ServicesDatabase	Data Storage Component	Holds service definitions, availability, and user bookings.
Profile Screen Interface		
ProfileScreen	UI Screen - Module	Displays and manages the user's personal information and account settings.
UserInfoSection	UI Component	Shows basic user details (name, email, disability type).
EditProfileButton	UI Control	Opens the edit profile interface.
SettingsPanel	UI Component	Contains app preferences and accessibility settings.
LogoutButton	UI Control	Logs the user out of the system.
UserProfileService	Backend Service	Handles profile updates and user data retrieval.
UsersDatabase	Data Storage Component	Stores user account information and editable profile data.

Table 26- Classification

8.2 Definition

8.2.1 LoginScreen interface

The LoginScreen interface enables pre-registered users to access their accounts within the Sanad system.

Its purpose is to verify login credentials (email and password) to ensure secure and reliable access to services.

Semantically, it represents the system's security gateway, ensuring that customized services, personal data, the forum, analytics, and the map are accessible only to authorized users.

It also serves as the transition point between the general user and the defined user within the system, making it a crucial component in achieving privacy and protecting the data of people with disabilities.

8.2.2 SignUpScreen interface

The SignUpScreen interface enables new users—whether individuals with disabilities, caregivers, or service providers—to register with the Sanad system.

Its purpose is to securely collect personal and login information to create a valid user account.

Semantically, it represents the official entry point to dedicated services, ensuring that only verified users can access support features, the community, services, and maps.

8.2.3 HomeScreen interface

The HomeScreen interface serves as the main navigation hub within the system.

Its purpose is to provide quick and direct access to essential services such as the map, services, analytics, forum, and profile.

Semantically, it represents the user's main control panel, connecting all subsystems into a single, user-friendly interface that complies with accessibility requirements.

8.2.4 ReportsAnalyticsScreen interface

The ReportsAnalyticsScreen interface provides users with analytics, summaries, and visual data based on system activity.

Its purpose is to display information that helps users understand their usage history, activity, and service patterns.

Semantically, it transforms raw data into actionable insights, supporting decision-making and improving the user experience.

8.2.5 MapScreen's interface

MapScreen's interface displays an interactive map focused on accessible locations.

Its purpose is to help users identify suitable places for navigation or use through filters, tags, and search features.

Semantically, it represents the system's geographic layer, linking user needs to real-world accessibility data.

8.2.6 ForumScreen interface

The ForumScreen interface provides an interactive community environment where users can share experiences, ask questions, and support one another.

Its purpose is to foster communication among users and facilitate knowledge exchange.

Semantically, it represents the social and supportive aspect of the system, fulfilling the needs of the community and its interaction.

8.2.7 ServicesScreen interface

The ServicesScreen interface displays a list of available services, such as booking sign language interpreters or requesting companions.

Its purpose is to enable the user to browse services and make bookings.

Semantically, it represents the service delivery layer that connects user needs with the resources available in the system.

8.2.8 ProfileScreen interface

The ProfileScreen interface allows users to manage their personal information, preferences, and account settings.

Its purpose is to ensure accurate data and allow for customization based on each user's disability needs.

Semantically, it represents the user's identity within the system and ensures customizability, privacy, and accessibility for people with disabilities.

8.3 Responsibilities

Component Interface	Main Responsibilities Behaviors	What the Component Accomplishes	Roles Played	Services Provided to Clients
LoginScreen	Authenticate pre-registered users, validate credentials, provide secure access	Grants authorized users access to the system	Security gateway, user verifier	Login validation, access to personalized services, privacy protection
SignUpScreen	Collect personal and login information securely, validate data	Registers new users with verified accounts	Registration facilitator, security verifier	Account creation, verification, access to support services
HomeScreen	Provide navigation to essential services, display dashboard	Central hub connecting all system features	Main control panel, navigation facilitator	Quick access to map, services, analytics, forum, profile
ReportsAnalyticsScreen	Display summaries, visual data, analytics	Helps users understand activity, usage history, service patterns	Data presenter, insight generator	Analytics, usage reports, visual insights for decision-making
MapScreen	Display interactive accessible map, allow filtering/search	Helps users identify accessible locations	Geographic navigator, information connector	Map visualization, filters, search, location info
ForumScreen	Enable posting/replying, support interaction	Fosters community communication and knowledge exchange	Community facilitator, moderator	Message posting, discussion, Q&A, support

ServicesScreen	Display available services, enable bookings	Allows users to browse and book services	Service connector, scheduler	Service browsing, booking requests, confirmations
ProfileScreen	Manage personal info, preferences, account settings	Ensures accurate data and customizable experience	Identity manager, customization facilitator	Profile management, privacy control, accessibility settings

Table 27– Responsibilities

8.4 Constraints

Component	Constranint
Login Screen	• The user must have a pre-registered account (a prerequisite). • The email and password fields cannot be left blank. • The password must match the stored encrypted version. • Verification must be completed within an acceptable response time. • The number of incorrect login attempts is limited for security reasons.
Sign Up Screen	• All other objects must be scanned before funds. • The email address must be in the correct format. • The password must match the one on the account. • Do not use duplicate email addresses (uniqueness is required). • Data must be stored securely in accordance with privacy regulations.
Home Screen	• Essential components (map, services, forum, profile) must load without delay. • Users must be logged in beforehand (prerequisite). • Only content relevant to the current user will be displayed.
Reports & Analytics Screen	• Relies on stored usage data. • Creating emergency winds requires accurate data. • Large amounts may not be displayed when reports (and their data) are shown. • Users can only access their own reports (with permissions).
Map Screen	• Requires an internet connection or map data. • Location data must be in a standard format. • Filtering operations must not exceed the system's processing capacity. • User location services require prior authorization (a prerequisite).
Forum Screen	• Only registered users can participate or reply. • Texts must not exceed the length limit. • Inappropriate content will be filtered (moderated). • Posts must prevent the storage of mobile data.

Services Screen	• Requires a list of services to connect to back-end APIs. • Must contain reservation requests for all required data. • Prevents two overlapping requests simultaneously (logical constraint). • Service availability depends on the service schedule (external constraint).
Profile Screen	• Users must log in to view or edit their profiles. • The input format (name, phone number, email address) must comply with verification rules. • Sensitive changes (email address, password) require re-verification. • Data must remain consistent across the system (static).

Table 28 –Constraints

8.5 Composition

8.5.1 Sign Up Screen

The registration interface consists of a set of sub-elements that work together to create a new user account:

- Full Name Field: Receives the username.
- Email Field: Receives the user's email address for verification and account creation.
- Password Field: For secure password entry.
- Confirm Password Field: Ensures passwords match.
- Sign Up Button: Executes the registration process.
- Validation Service: Verifies the entered information.
- UserAccountService: Creates the account and stores the data.

8.5.2 Login Screen

The login interface consists of the following elements:

- Advanced Email Location (Email field): Various forms of user identification.
- Password Search (Password field): For secure password entry.
- Login Button (Login button): Submits login credentials.
- Go to Signup Button: Takes the user to the registration interface.
- Authentication Service: Verifies the validity of login credentials.

8.5.3 Home Screen

The home screen consists of the following elements:

- Main navigation bar: Provides quick access to essential pages.
- Accessibility area (feature icons): For each service or system component.
- User summary: Displays the current account information.

8.5.4 Map Screen

The planet map consists of the following elements:

- Interactive Map: Displays the current location.
- Location Markers: Represent the Facebook map.
- Filter Options: Allows you to filter options as needed.
- Location Service: Retrieves the user's current location.

8.5.5 Reports & Analytics Screen

It consists of:

- Statistics Box (Statistics Cards): Summarizes basic information.
- Atmospheric Effects (Graphs): Displays interaction and usage.
- Analytics Engine: Transforms data into visual results.
- Log Database: Stores previous user data.

8.5.6 Forum Screen

It consists of:

- Posts List: Displays community posts.
- Post Editor: A new post editor.
- Replies Area: For interacting with posts.
- Moderation Service: For content filtering.

8.5.7 Services Screen

It consists of:

- Services List: Displays available services.
- Service Details: Requirements and details for each service.
- Booking Button: Request the service.
- Services API: Data generated by the front-end system.

8.5.8 Profile Screen

It consists of:

- User Information (User Information Fields): such as name, email, and phone number.
- Preferences Settings (Preferences): to configure the user's basic requirements.
- Edit Data Button (Update Button): A nicer option.
- Account Management Service (Account Management Service): Provides updates.

8.6 Uses/Interactions

8.6.1 User Management & Authentication Subsystem

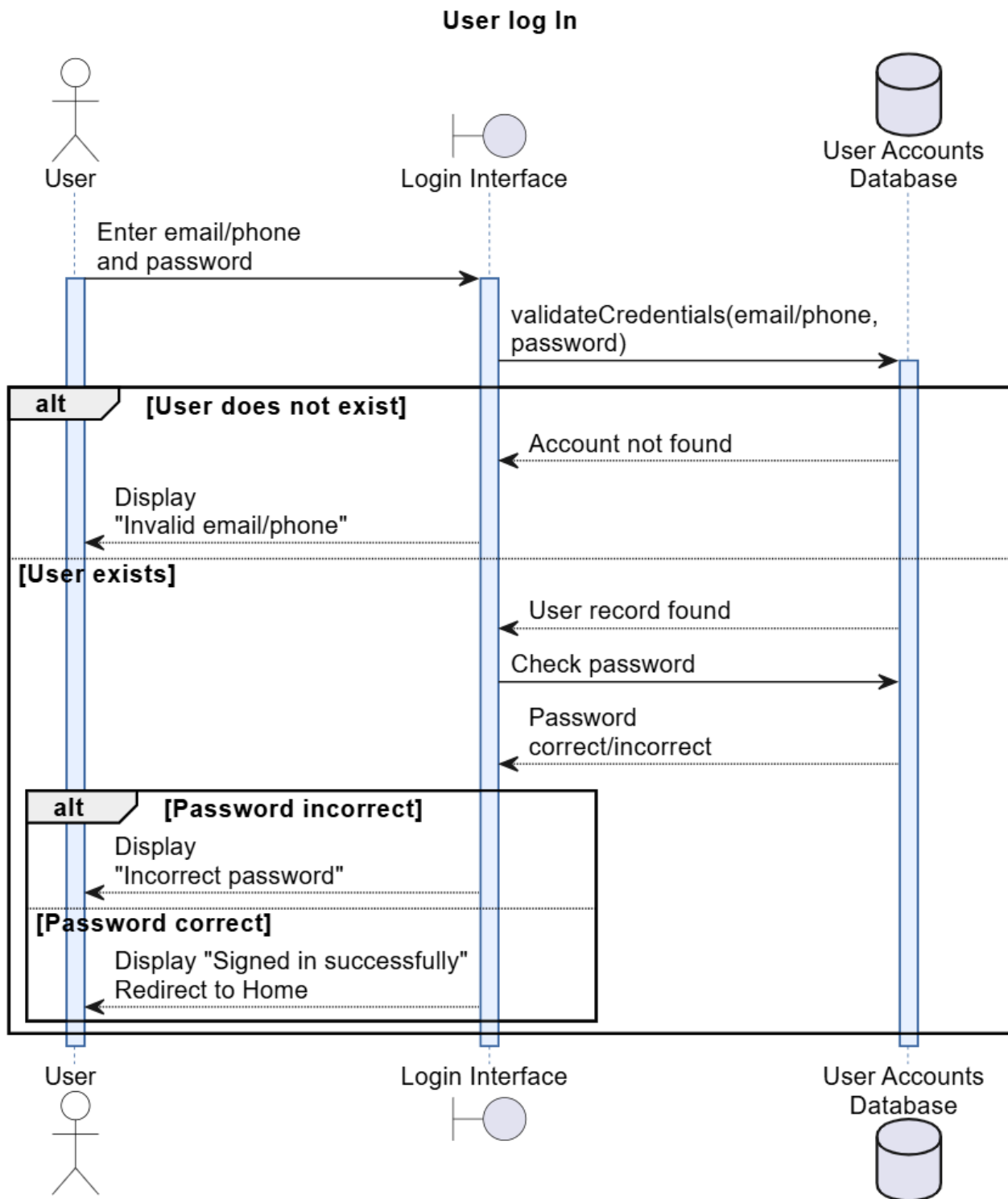


Figure 23– User Management & Authentication Subsystem

8.6.2 Sign Up & User Profile Initialization

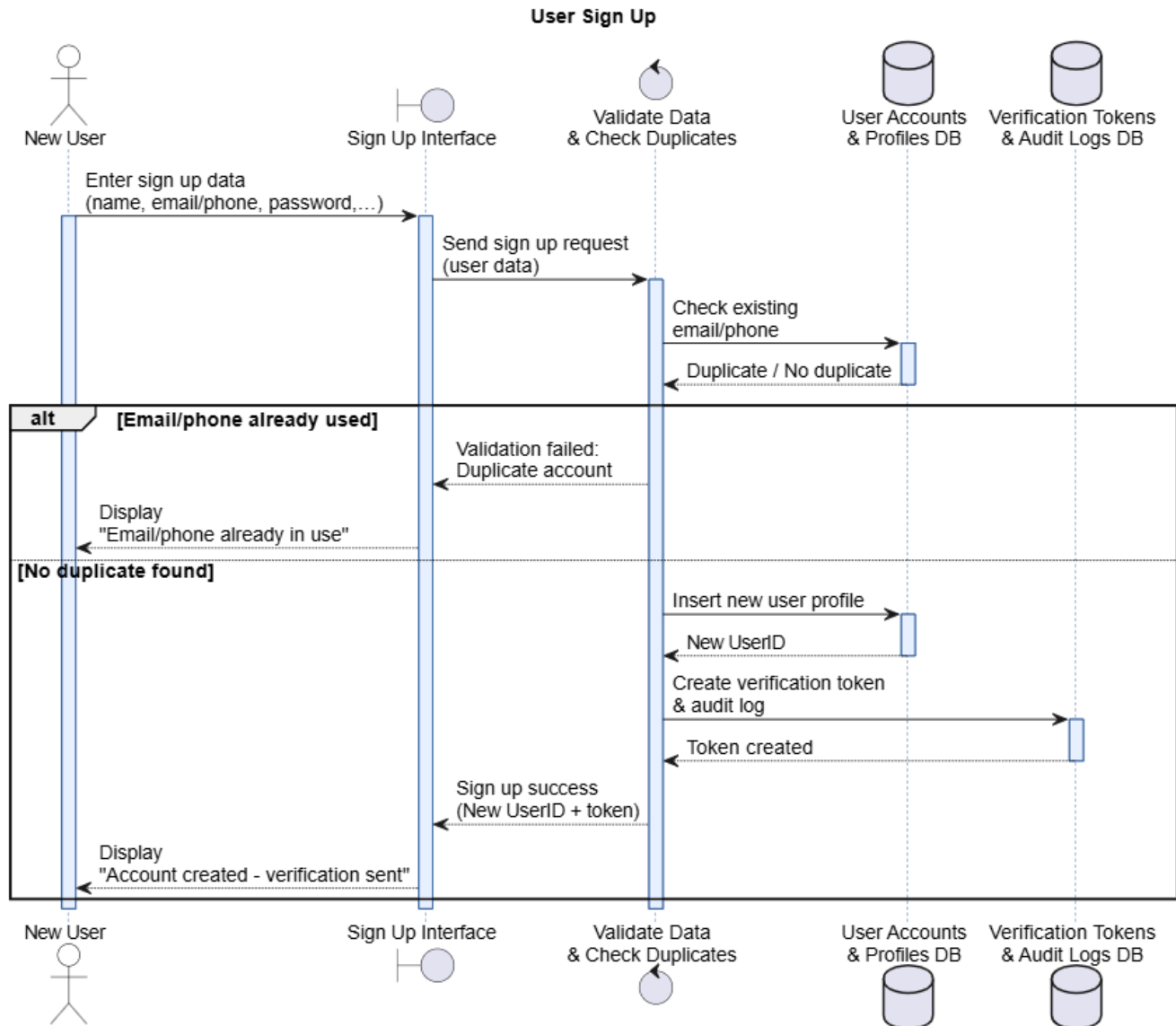


Figure 24– Sign Up & User Profile Initialization

8.6.3 Home Dashboard & Personalization

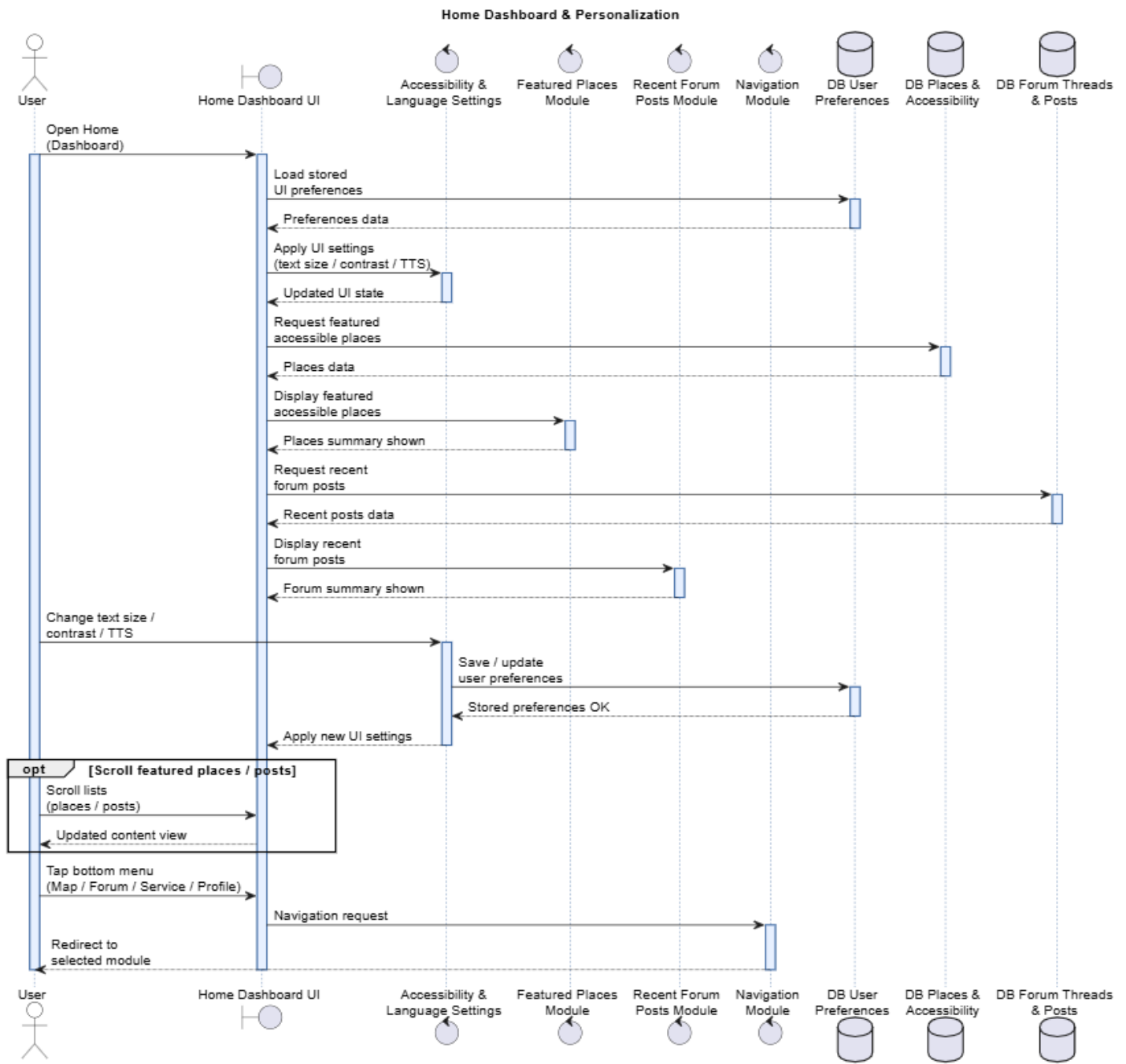


Figure 25–Home Dashboard & Personalization

8.6.4 Map & Accessible Places Subsystem

8.6.4.1 Search & Filter Places

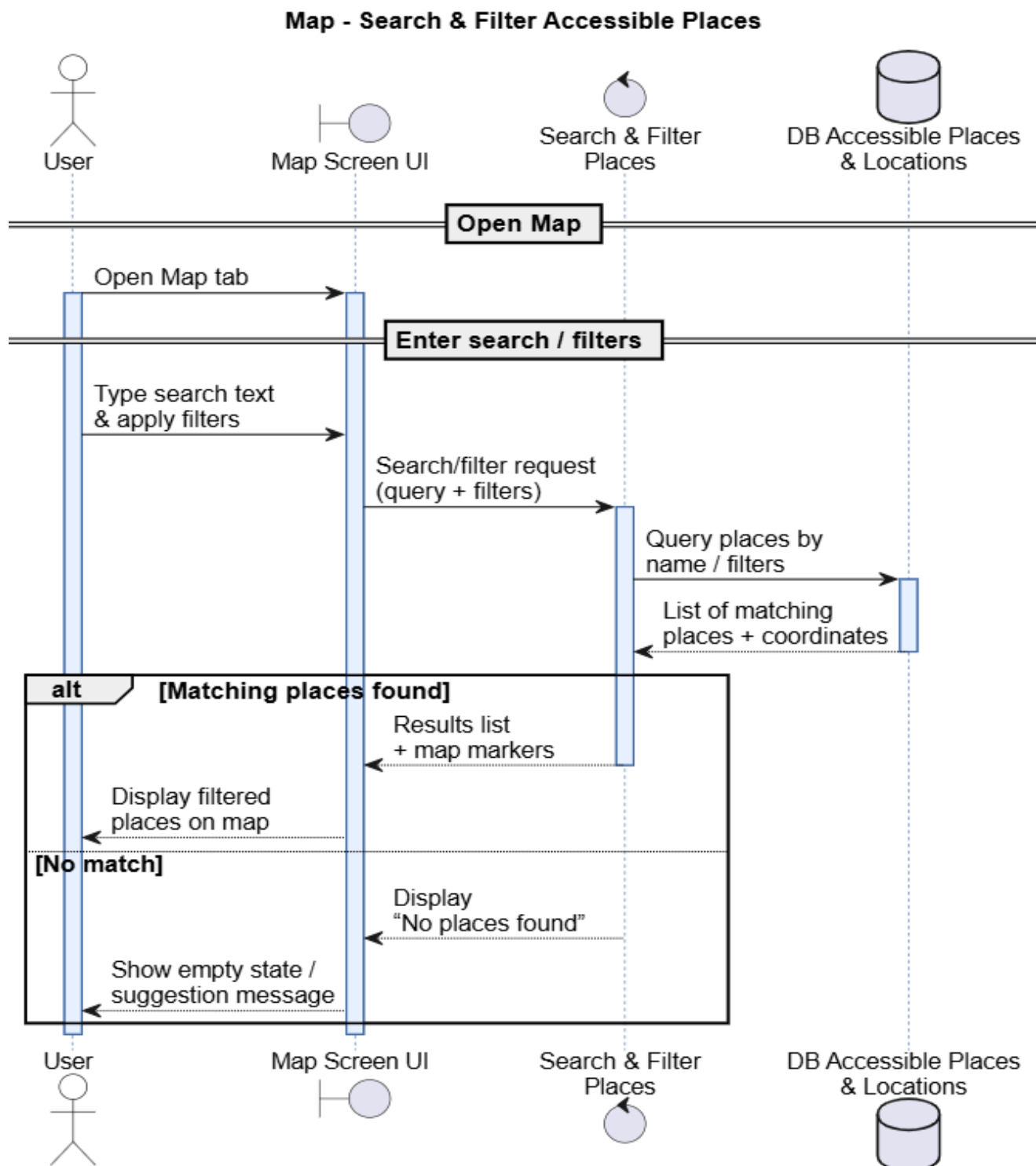


Figure 26-Search & Filter Places

8.6.4.2 View Place Marker & Details

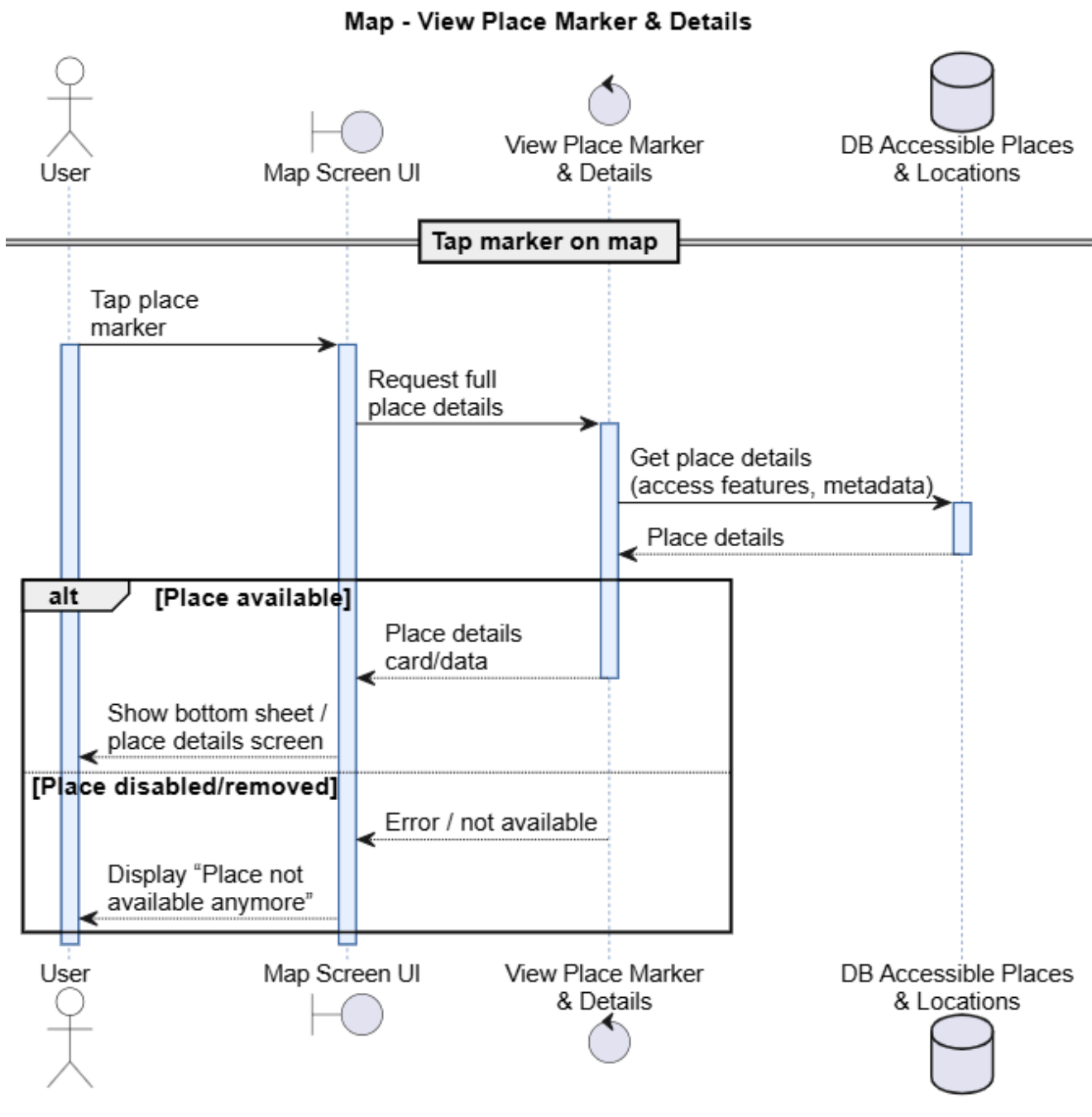


Figure 27- View Place Marker & Details

8.6.4.3 Add / Suggest New Place

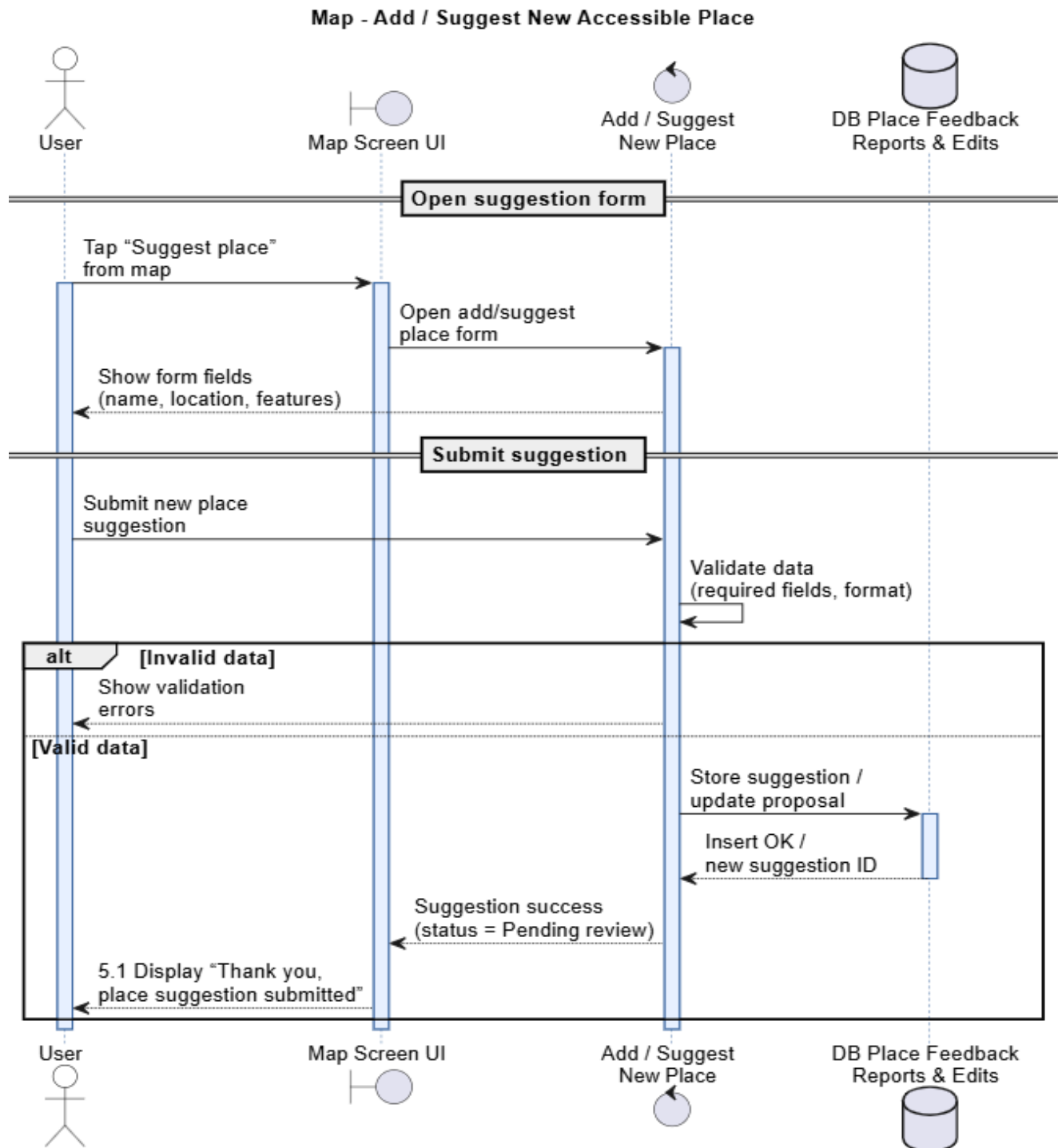


Figure 28-Add / Suggest New Place

8.6.4.4 Adjust Map Layers & Info

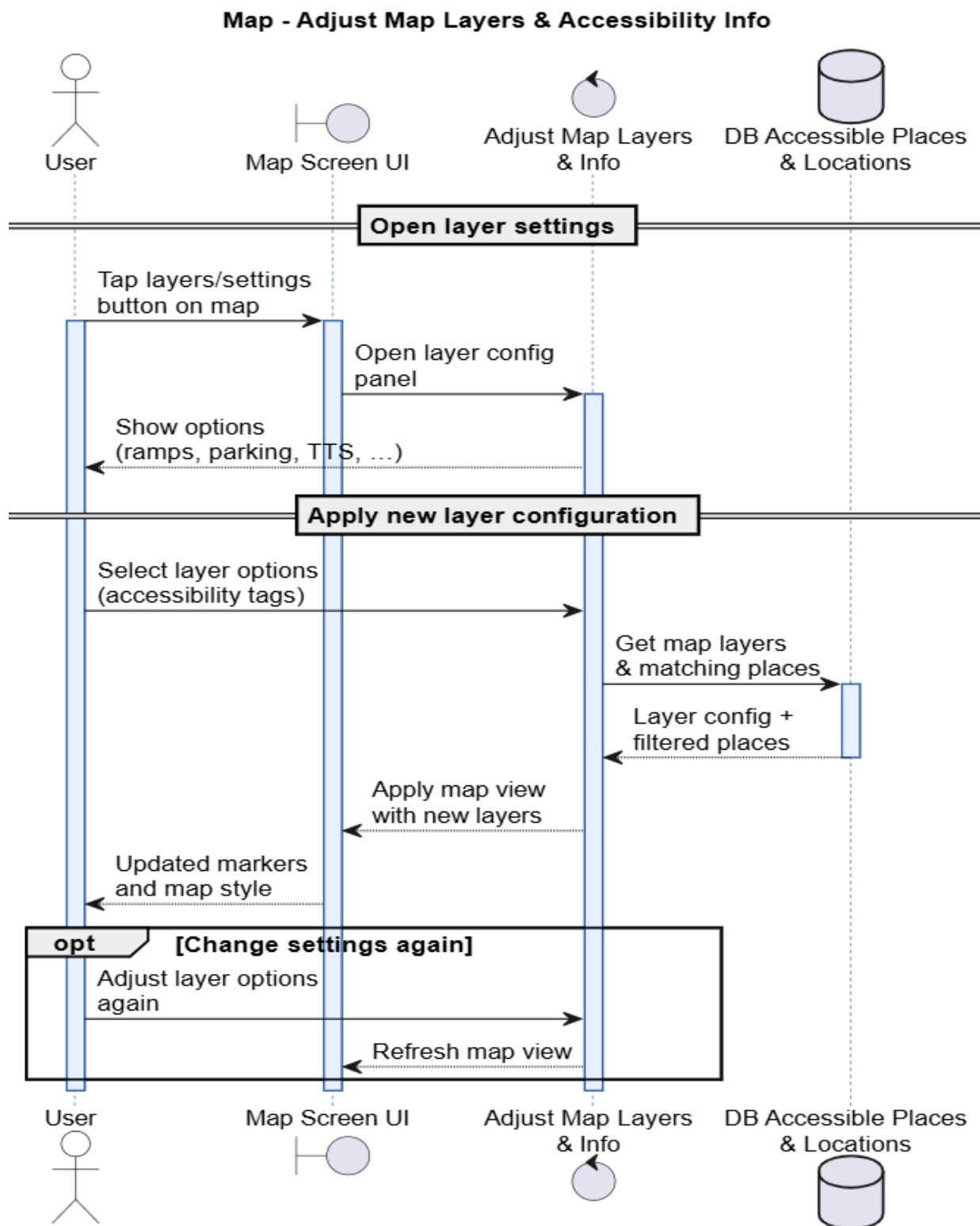


Figure 29 – Adjust Map Layers & Info

8.6.5 Community & Forum Subsystem

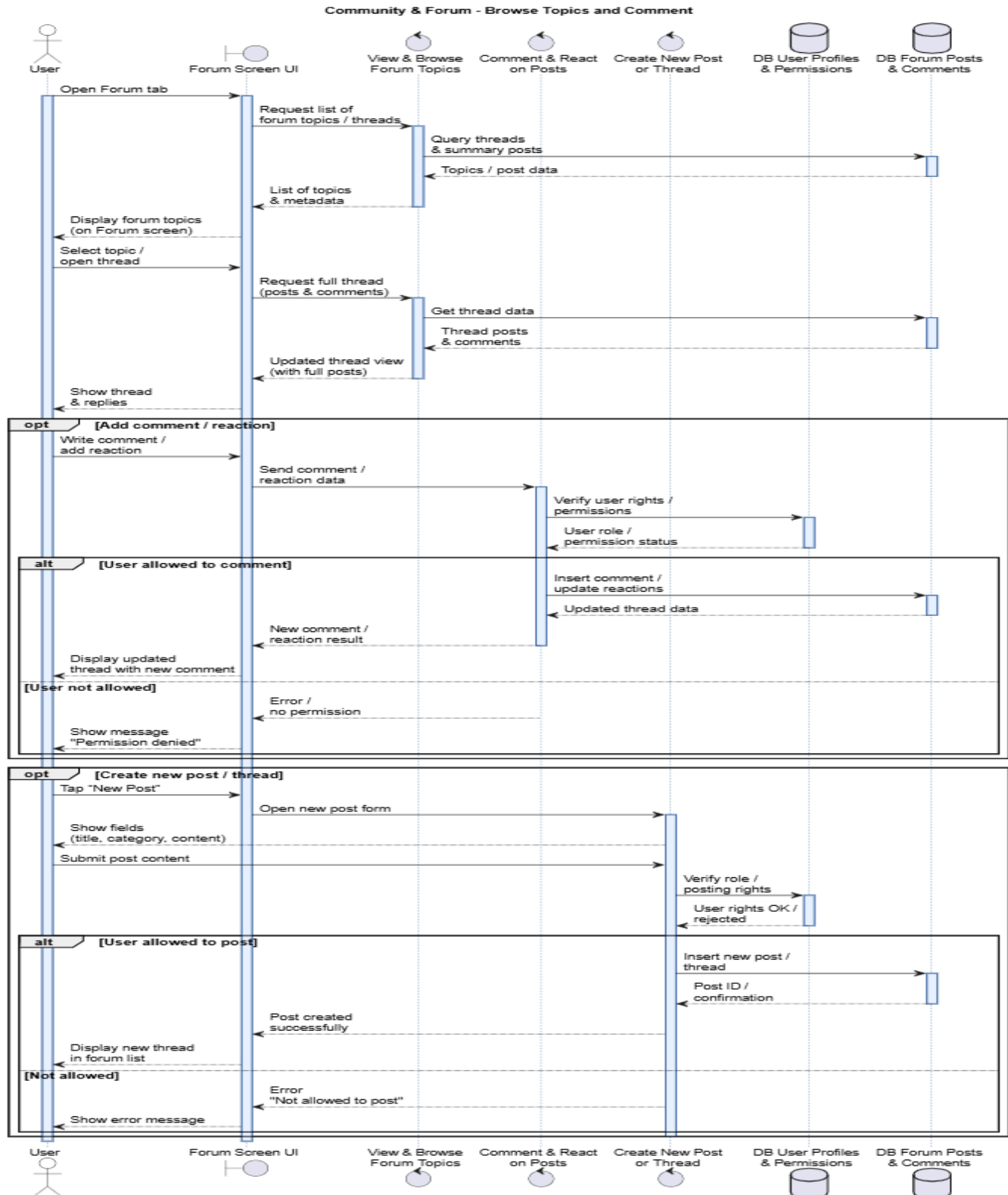


Figure 30- Community & Forum Subsystem

8.6.6 Service & Appointment Booking Subsystem

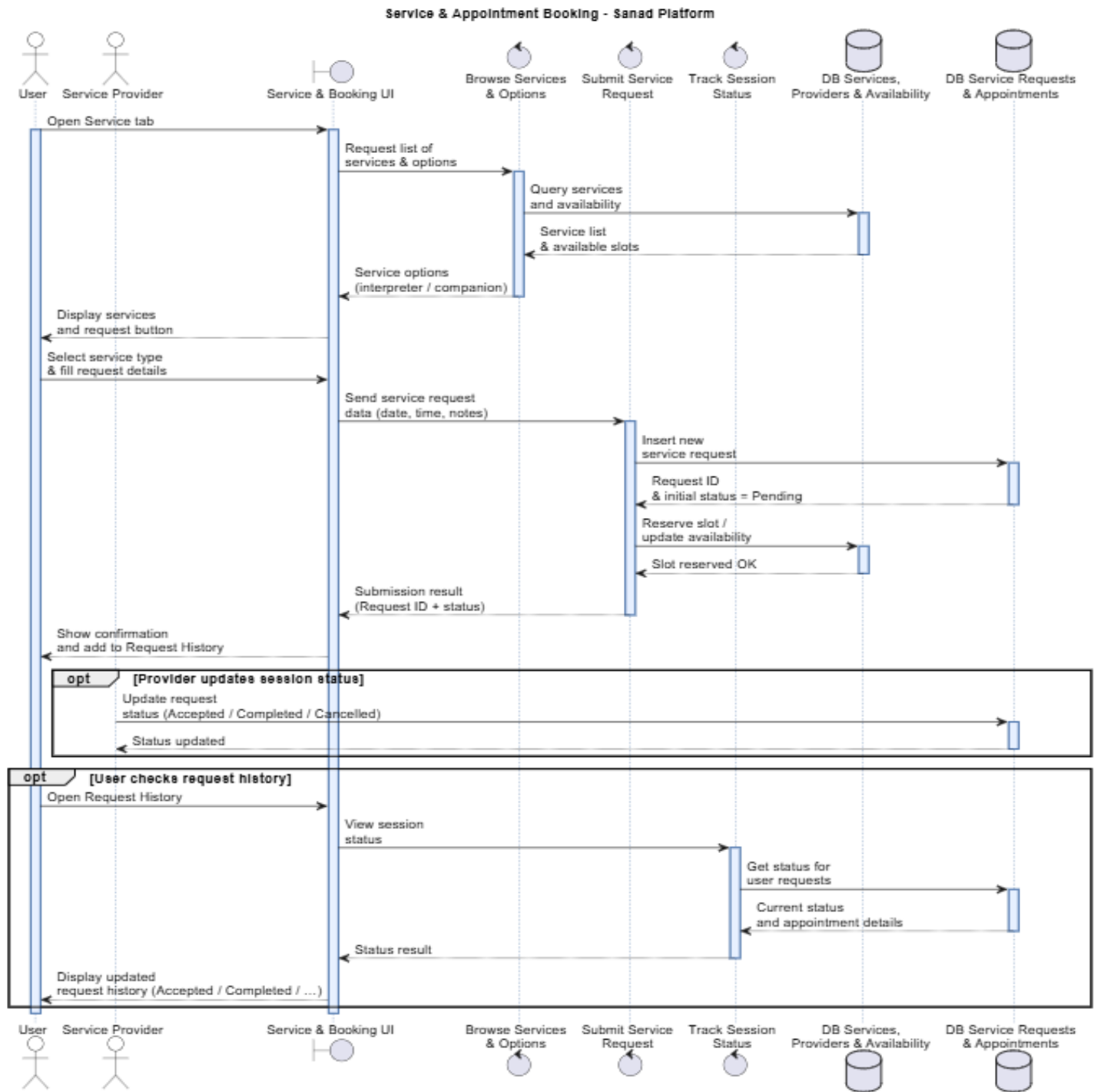


Figure 31 – Service & Appointment Booking Subsystem

8.6.7 Profile & Accessibility Preferences

8.6.7.1 View Profile Summary

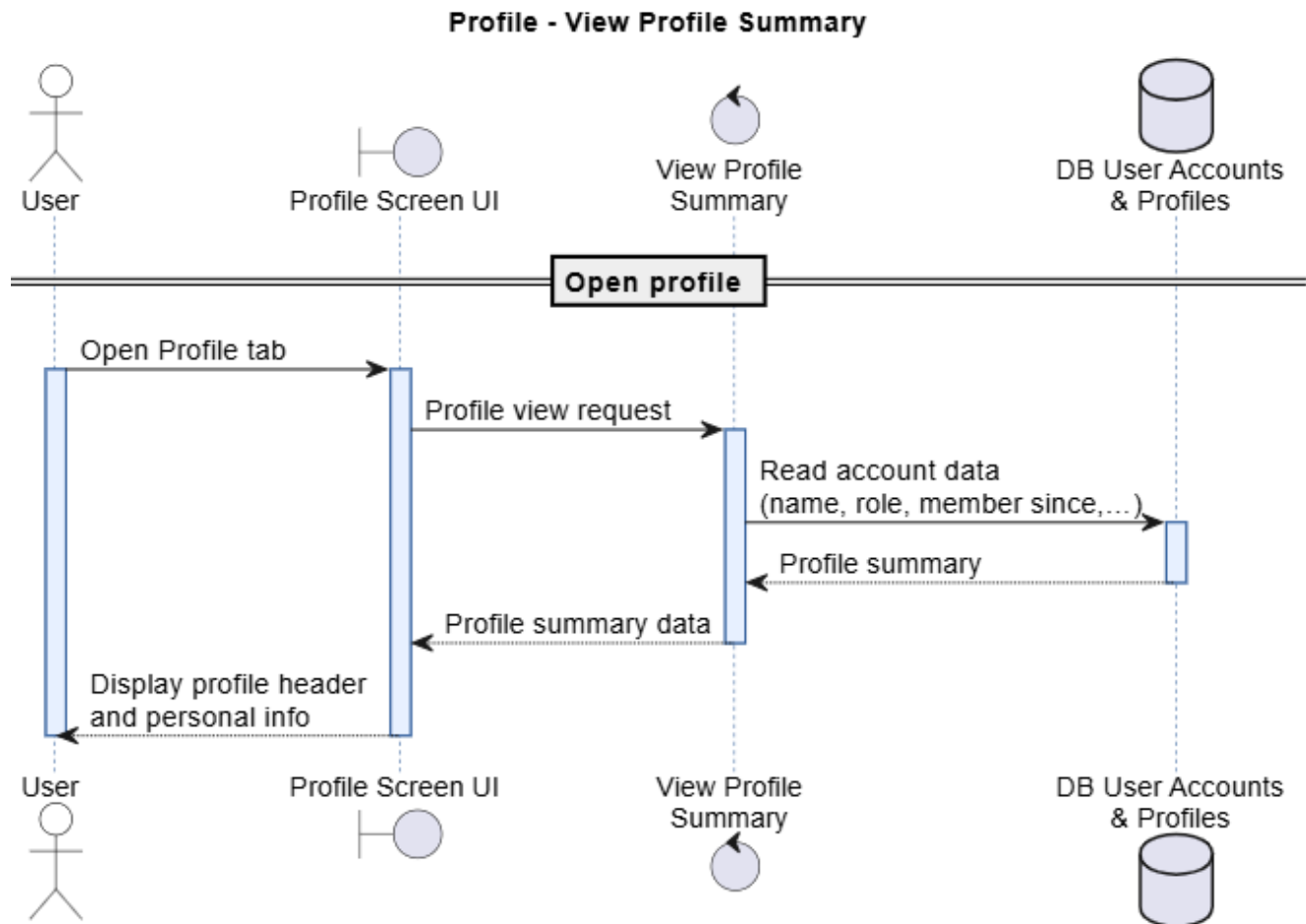


Figure 32- View Profile Summary

8.6.7.2 Edit Personal Information

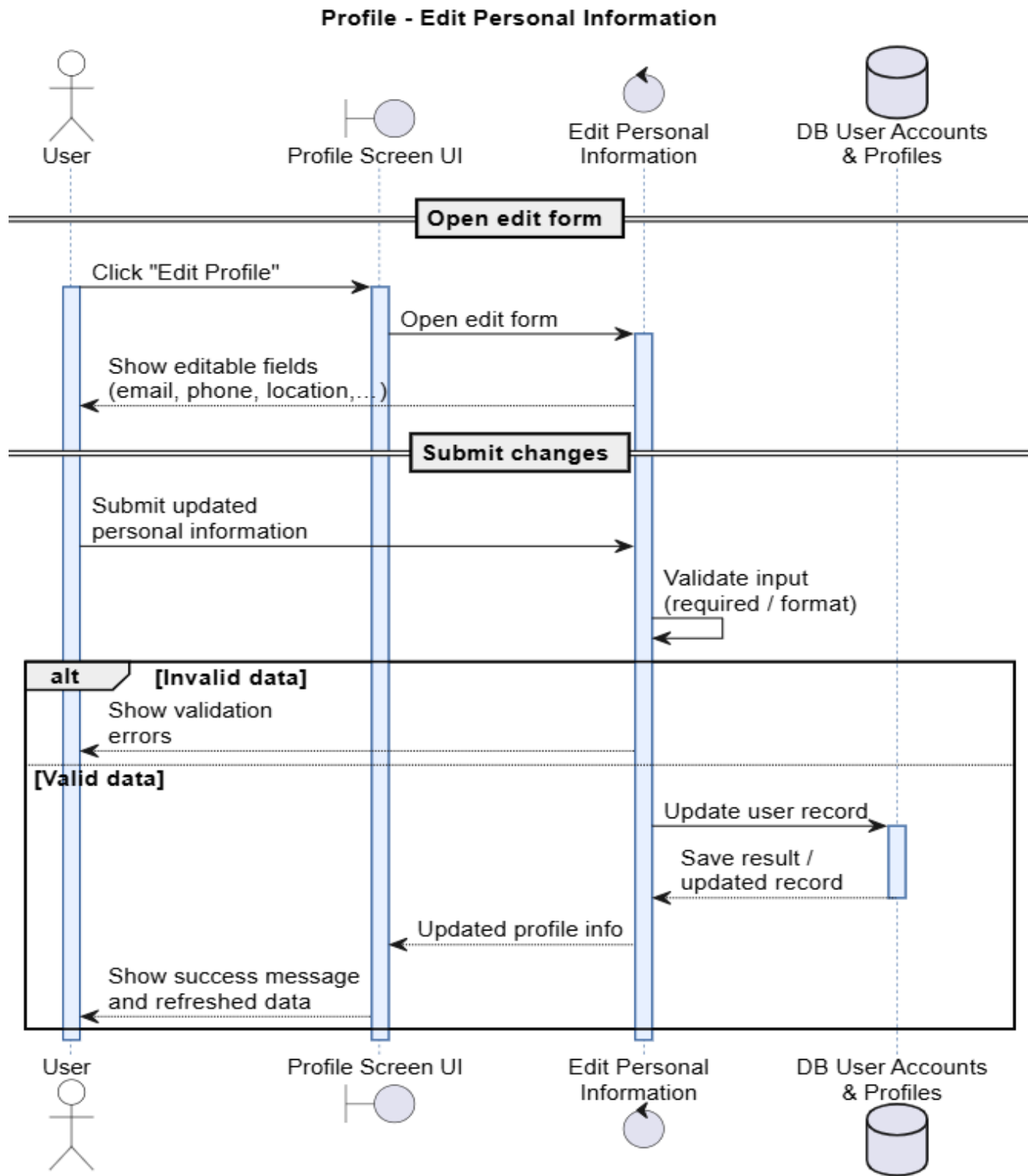


Figure 33- Edit Personal Information

8.6.7.3 Update Accessibility Preferences

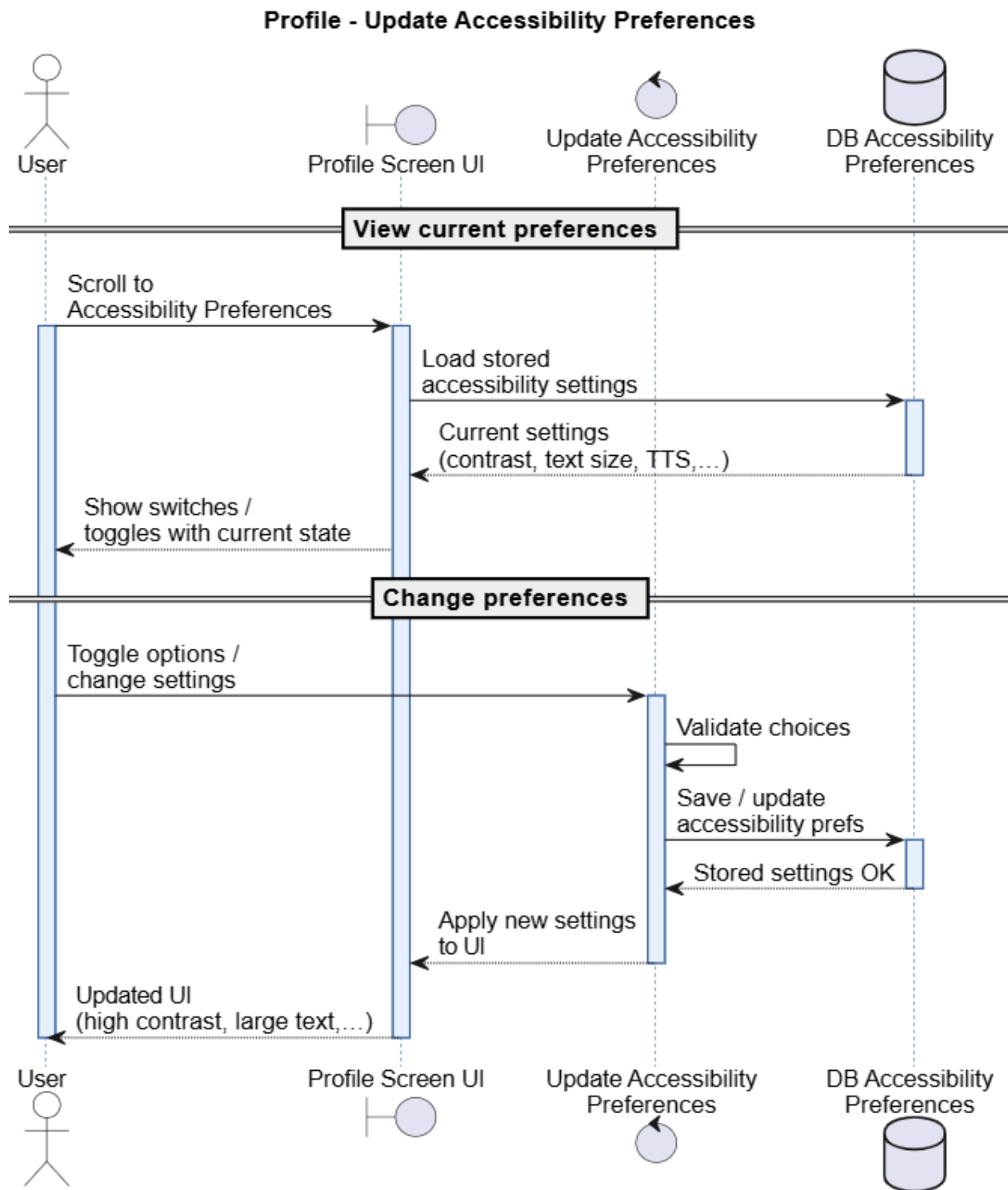


Figure 34- Update Accessibility Preferences

8.6.7.4 View Service / Support History

Profile - View Service / Support History

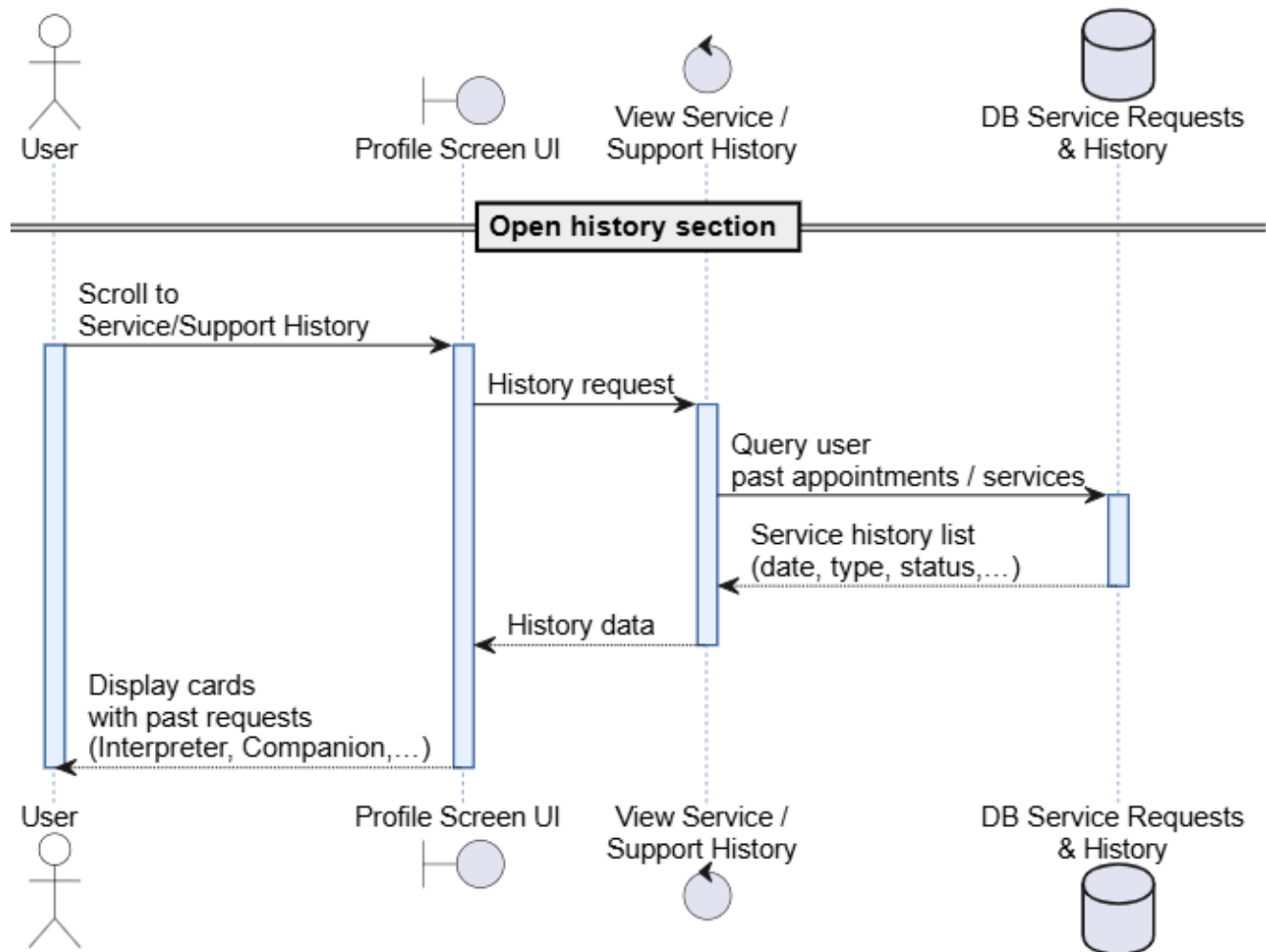


Figure 35- View Service / Support History

8.7 Resources

8.7.1 Hardware Resources

Resource Category	Description	Specification / Requirement
Application / Web Servers	Runs backend services, APIs, authentication, booking and forum.	• Multi-core CPU (Intel i5 or higher) • 8–16 GB RAM • 200 GB SSD • ≥ 100 Mbps network bandwidth
Database Servers	Hosts user accounts, places metadata, appointments and forum data.	• 8 GB “RAM” • 200 GB SSD • Replica / backup server • Automatic daily backups
Backup Storage	Disaster recovery for DB and logs.	• Off-site or cloud storage • Supports incremental & full backups
Client Devices	Devices used by users	• iOS 12 / Android 10 • Windows 10/11 & macOS browsers • Supports accessibility features (screen reader and contrast)

Table 29- Hardware Resources

8.7.2 Software & Platform Resources

Category	Description	Tools / Technologies
Backend Platform	Core server logic and API layer	Node.js and Express.js, JWT auth and input validation
Frontend Platform	Web & mobile UI	React (Web) and Flutter / React Native (Mobile)
Database	Stores system entities	MySQL relational DB
Storage	File & media storage	Cloud storage for images (places, posts and assistive tech items)
Security Infrastructure	Protects user data & sessions	HTTPS/TLS, hashing, encryption, logging and monitoring systems
Admin / Monitoring Tools	System-level supervision	Log monitoring, performance dashboards and admin moderation console

Table 30-Software & Platform Resources

8.7.3 Data & Storage Resources

Resource Type	Description	Examples
Relational Data	Structured data stored in MySQL	Users, Profiles, Places, Accessibility Tags, Bookings, Forum Posts, Jobs and Courses
Indexes	Improve performance	Index on: email/phone, coordinates, post timestamps and appointment status
Backups	Protect against data loss	Daily automated backups and backup replication
Logs	Track activity & moderation	Login logs, content reports and accessibility preference changes

Table 31- Data & Storage Resources

8.7.4 External Services & APIs

External Service	Purpose in Sanad Platform	Notes / Requirements
Mapping API (Google Maps)	Map rendering, coordinates and geocoding	Requires API key & quota management
Video Call API	Interpreter sessions	Must support encrypted communication
SMS / Email Services	OTP, notifications and booking reminders	Stable delivery & regulatory compliance
Employment / Learning APIs	Fetch job listings & training courses	Integrated in Learning & Employment subsystem
Optional Payment Gateway	For donations or paid services	Must meet PCI-DSS & local rules

Table 32-External Services & APIs

8.7.5 Human & Operational Resources

Role	Responsibilities	Notes
System Administrator	Deployment, backups, monitoring and performance	Handles server health & infrastructure
Moderators	Review places, posts, reports and updates	Ensures safety & content quality
Support Staff	Assist users with issues or accessibility problems	Email & helpdesk-based
Developers / DevOps	Maintain code, CI/CD, fixes and updates	Works on continuous improvement
Accessibility Specialists	Validate WCAG 2.1 compliance	Ensures usability for all disabilities

Table 33-Human & Operational Resources

8.7.6 Concurrency, Race Conditions & Deadlock Handling

Issue Type	Possible Scenario in Sanad Platform	Risk	Resolution Method
Race Condition	Two users try to book the same interpreter appointment slot at the same time.	Double booking / inconsistent state	Use transactional locking in DB and atomic writes.
	User updates accessibility preferences while device syncing occurs.	Preferences overwritten	Use last write wins timestamping and version control.
	Multiple moderators edit the same forum post simultaneously.	Data loss or conflicting edits	Apply optimistic concurrency control using row version numbers.
Deadlock	Two DB transactions: one updating user profile and another updating session tokens.	DB halt / timeouts	Use short lived transactions, consistent locking order and timeout-based deadlock detection.
	Multiple booking operations competing for provider availability rows.	Scheduler stop	Use fine grained locking and retry logic.

Table 34- Concurrency, Race Conditions & Deadlock Handling

8.8 Processing

All components of the Sanad Platform including: their inputs, outputs, internal processing logic and any exceptions or abnormal conditions are documented in the Sanad Platform Software Requirements Specification (SRS) as sections 3.2. This has been identified as the main reference for processing details, behavioral flows and design decisions in the writing of the Detailed System Design (SDS).

It includes:

- inputs and outputs,
- internal algorithms,
- changes of system state,
- concurrency considerations,
- initialization and cleanup routines, and
- handling of exceptional or abnormal conditions.

Each subsection maps directly to the major functional modules defined in the SRS.

8.9 Interface/Exports

The range of services exported by each subsystem of the Sanad Platform consists of the resources, data structures, types, constants, operational routines, and exception conditions, which are available for use by other portions of the system. Although the exact definitions of these service elements are not repeated here, their descriptions are pointed out from previous subsections of this Detailed System Design (SDS), where the attributes of each component which are the Classification, Definition, Responsibilities, Constraints, Composition, Uses, Resources, Processing and Interface have been explained thoroughly.

Application developers working on the Sanad Platform must have an explicit understanding of all the subsystem functions and functionality interactions in order to ensure that the system operates

seamlessly. Therefore, it is necessary to consider the component's inputs, outputs, pre condition, post condition and constraints when exported interfaces are defined, as introduced in Sections 8.1 and 8.2. Similarly, Sections 8.3 through 8.7 used detailed sequence diagrams and flowcharts to articulate the behavioral flows, processing logic, and interaction sequences of each subsystem. These diagrams can be viewed as the main reference points for understanding how each component interacts with the other subsystems and what it exposes to the external world.

As with common practices in general software design documentation, this section does not attempt to re-create low-level implementation particulars that would better belong directly in the annotated source code. For example, much of the interface definitions (method signatures, parameter types, returns values, specified possible exceptions) will be contained in the specific method definitions within the respective sources. Developers are strongly encouraged to maintain source code comments and annotate diagrams as implementations evolve. Overall, this subsection consists mostly of references to the previously outlined diagrams and component descriptions, not a restating of source level details.

8.10 Detailed Subsystem Design

The user-oriented subsystem of the Sanad platform consists of several interconnected components that work together to provide support services for people with disabilities. Each component is designed as an independent unit with a specific purpose. The subsystem comprises eight main units: Login, Registration, Homepage, Map, Reports and Analytics, Forum, Services, and Profile.

1. Login Module

This module handles the user identity verification process. It includes input fields, verification logic, and secure interaction with the backend authentication interface. It ensures the validity of login data, error handling, and access control before granting the user access.

2. Sign Up Module

This module is responsible for creating new accounts. It includes input fields, password matching logic, email verification, duplicate detection, and secure submission of the user account service. Its internal workflow ensures data integrity and security.

3. Home Module

This module serves as the primary navigation interface. It contains links to core modules and displays dynamic content related to the current user. It includes interface elements such as icons and summary cards, as well as permission control rules.

4. Map Module

This module integrates an interactive map supporting predefined locations. It includes location services, filtering tools, displaying markers on the map, and data synchronization with the backend system. The control workflow manages map loading, filter application, and location display.

5. Reports & Analytics Module

This module processes stored data and transforms it into visual insights. It includes a data fetch engine, graphing logic, statistics calculations, and mechanisms for handling incomplete or incorrect data.

6. Forum Module

This module enables user interaction. It includes post handlers, reply managers, content filters, and API connection layers. The system's behavior ensures content integrity and safe, moderate interaction.

7. Services Module

This module displays available services and handles bookings. It includes data fetchers, booking handlers, form validation, and synchronization with service provider schedules. It maintains consistency and prevents booking conflicts.

8. Profile Module

This module handles user personal data. It includes editable fields, preference handlers, change validation logic, and secure communication with backend update services. It ensures data privacy and accuracy.

9. Other Design Features

All design features of the **Sanad Platform** have been comprehensively detailed in the preceding Sections and Subsections of this Software Design Document. The presented content fully encompasses the architectural, functional, and interface specifications required to support the platform's operational objectives and ensure consistency across all system components.

10. Requirements Traceability Matrix

SRS Requirement ID	Technical Assumption(s) / Customer Need(s)	Functional Requirement Summary	User	Association ID in SDS	System Component(s)
FR1.1	Users need secure account creation	Create account using email/phone & password	User	7.1.1 Sign Up	Authentication Service, User Management
FR1.2	Users require accessible interface	Specify disability type	User	7.2.6 Manage Profile & Accessibility Preferences	Accessibility Engine, Profile Manager
FR1.3	Secure access required	Log in using registered data	User	7.1.2 Sign In	Authentication Service
FR1.4	Users must manage their data	Update profile & change password	User	7.2.6 Manage Profile	Profile Manager
FR-2.1	Users need map-based search	Map search + filters + list/map view	User	7.2.2 View Accessible Map, 7.2.1 Manage Home	Map Engine, Search/Filter Module
FR-2.2	Users require details before visiting	Show place details	User	7.2.2 View Accessible Map, 7.2.1 Manage Home	Place Details Module
FR-2.3	Users share ratings	Ratings and reviews submission	User	7.2.1 Manage Home	Reviews Module
FR-2.4	Moderation ensures data accuracy	Moderation for new places/reviews	Admin	7.3.2 Moderate Forum Content	Moderation Tools
FR-2.5	Users need safety/accuracy	Report inaccurate content	User	7.2.x Related to map & posts	Reporting Module

FR-3.1	Need a communication space	Forum with categorized topics	User	7.2.5 Manage Forum	Forum Module
FR-3.2	Need safe community	Report inappropriate content	User	7.2.5 Manage Forum	Reporting Engine
FR-3.3	Stories require moderation	Inspirational stories section	User	7.3.2 Moderate Forum Content	Moderation Tools
FR-3.4	Engagement requires alerts	Notifications for replies	User	7.2.5 Manage Forum	Notification Service
FR-4.1	Access to interpreter is required	Book a sign-language interpreter	User	7.2.3 Manage Services Request	Booking Module
FR-4.2	Privacy is required	Encrypted video call	User	7.2.3 Manage Services Request	Video Call Engine
FR-4.3	Need companion requests	Request volunteer escort	User	7.2.3 Manage Services Request	Volunteer Request Module
FR-4.4	Need proper matching	Match volunteers by location	User	7.2.3 Manage Services Request	Matching Engine
FR-4.5	Scheduling flexibility needed	Reschedule appointments	User	7.2.3 Manage Services Request	Scheduling Module
FR-4.6	Ability to cancel	Cancel appointments	User	7.2.3 Manage Services Request	Scheduling Module
FR-4.7	Need appointment visibility	View appointment status	User	7.2.3 Manage Services Request	Status Tracking Module
FR-5.1	Users seek learning	View learning opportunities	User	7.2.4 Browse Education & Jobs	Education Module
FR-5.2	Need details clarity	View course details	User	7.2.4 Browse	Education Module

				Education & Jobs	
FR-5.3	Job search is required	Browse jobs + filters	User	7.2.4 Browse Education & Jobs	Jobs Module
FR-5.4	Need job transparency	Job details	User	7.2.4 Browse Education & Jobs	Jobs Module
FR-5.5	Need assistive tech info	Assistive tech library	User	7.2.4 Browse Education & Jobs	Assistive Tech Module
FR-5.6	Bookmarking improves usability	Save items (favorites)	User	7.2.4 Browse Education & Jobs	Favorites Manager
FR-5.7	External access needed	Open external link	User	7.2.4 Browse Education & Jobs	External Link Handler
FR-6.1	Admins must review content	Admin dashboard for reviewing additions	Admin	7.3.1 View Reports & Analytics	Admin Dashboard
FR-6.2	Maintain platform safety	Monitor forum + take action	Admin	7.3.2 Moderate Forum Content	Moderation Engine
FR-6.3	Need insights/reporting	Performance/Analytics reports	Admin	7.3.1 View Reports & Analytics	Analytics Engine
FR-6.4	Need control over staff	Manage moderator/volunteer accounts	Admin	7.3.3 Manage User Accounts	Admin Account Manager